



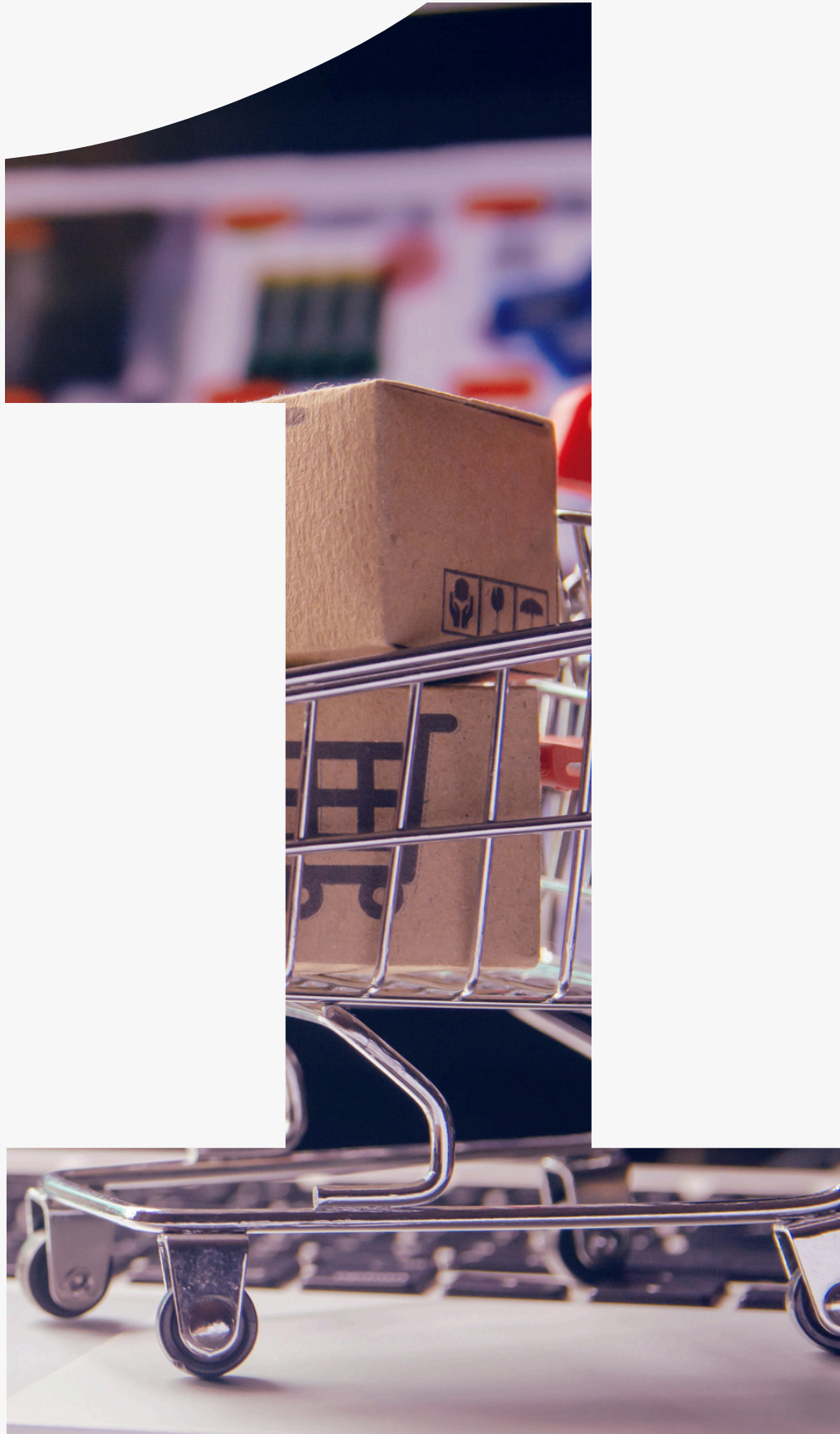
Reducing Churn at Scale:

A PREDICTIVE APPROACH FOR HIGH-IMPACT MARKETING DECISIONS

November, 2025

Tables of Content

Context & Business	1	
Leading context that explains the current situation and importance of the churn		
		EDA
		14
Data exploration, analysis of pattern and key findings related to churn		
Modelling	27	
Development of machine learning model based on evaluation metrics to predict churn of customer		
		Conclusion
		39
Summary of models and key findings throughout analysis influencing the churn behaviour and prediction		
Recommendation	42	
Strategic action to implement on reducing customer churn based on data driven decision and improvement of the model		



Context & Business

+



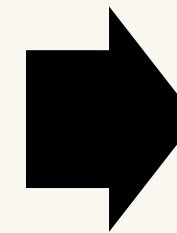


Background

Alpha Company

Product

Shopping Experience



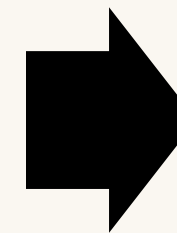
Medium Sized Online
Retail Business

Electronic devices,
gadget, and digital
lifestyle products

Fast, personalized, and
integrated experience

Challenge

High Churn Rate



Bad Profitability and
Sustainability

Why Company Churn Needs Immediate Attention ?

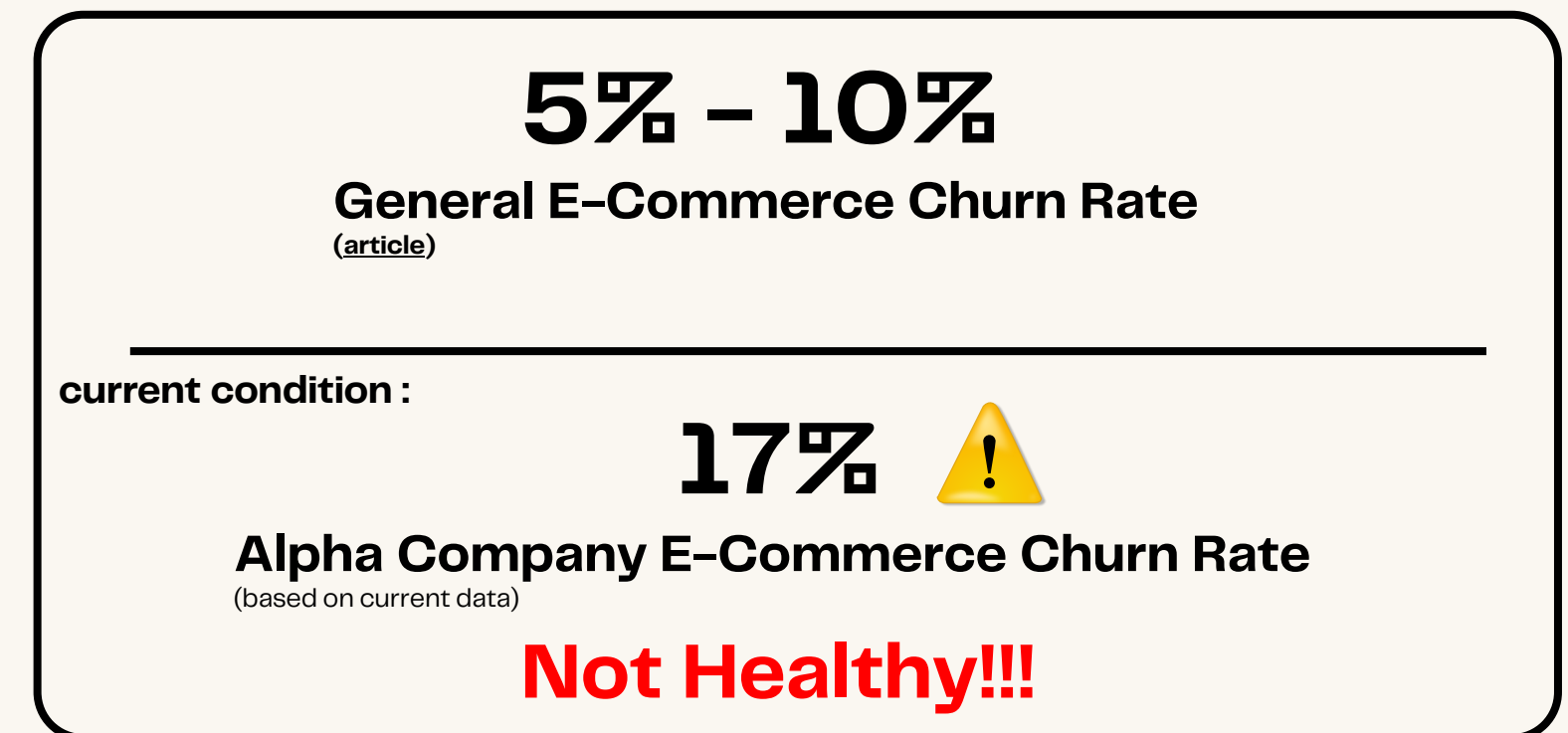
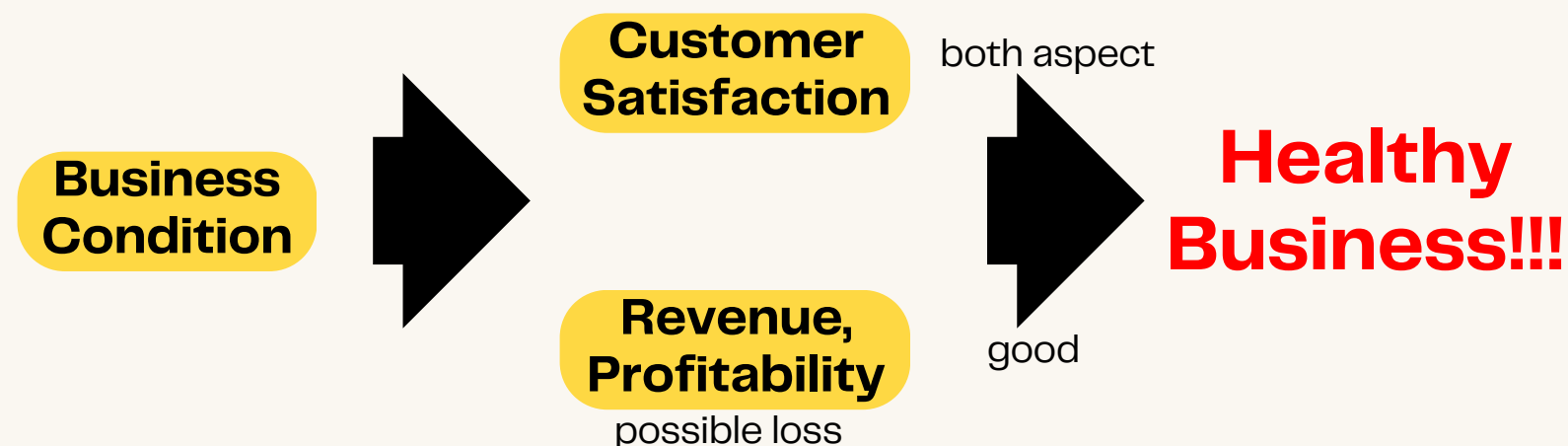
Churn

A condition where the customers stop using the services or the product.

in Alpha Company (online retail):

Customer **has not made/stop** any new purchases within a certain period of time.

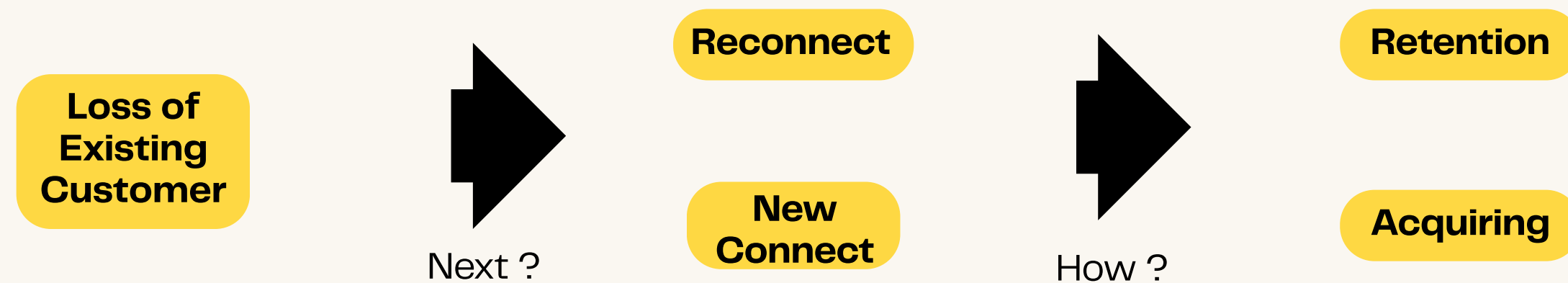
Churn indicates what?



"Understanding why customers leave is the first step toward building a strong and profitable online business"

Why Company Churn Needs Immediate Attention ?

How to Handle?



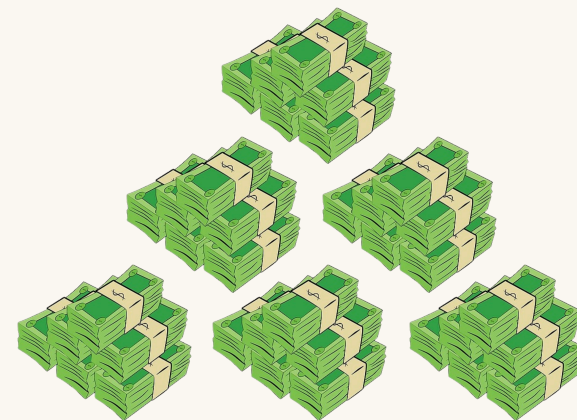
Which one to choose ?

**Minimize cost,
Maximize profit**

"Understanding why customers leave is the first step toward building a strong and profitable online business"

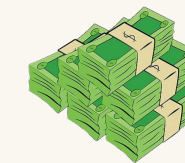
Why Retaining Customer is More Important ?

Acquiring

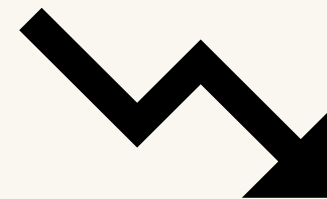


80\$
/customer

Retaining



20\$
/customer



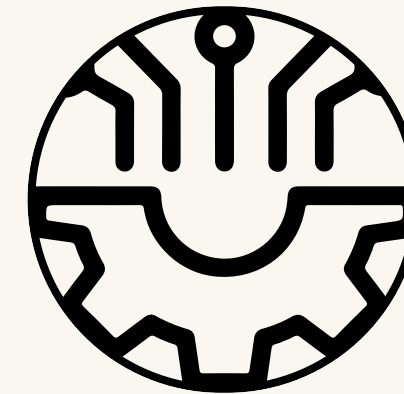
the cost of **retaining a customer** can save up to **3 – 6 times** the cost of **acquiring a new customer**
based on [\(article\)](#)

Retaining 5% of customer can increase the profit by 25% – 95% [\(article\)](#)

Problem Statement



What are the behavioral patterns and customer attributes that drives most customer to churn ?



How can machine learning models be developed to estimate churn risk and support business teams in executing timely, targeted retention strategies?

End Goals

1

Identifying churn patterns

Analyze behavioral variables such as tenure, device preferences, and payment methods, etc. to identify customers at risk of churn.

2

Increase personalization

Leverage customer behavioral data to **design** relevant **shopping experiences** and **increase** long-term **engagement**.

3

Improving the effectiveness of detecting customers churn

Implementing machine learning models with a focus on **capturing** as many potential **churning customers** as possible, so that high-risk customers are not missed in the retention strategy.

4

Optimizing customer retention cost

Using model prediction results to **focus retention** efforts only on customers most likely to churn, making **incentive allocation** and loyalty programs more **efficient** without sacrificing retention opportunities.

Stakeholder



**Chief Marketing
Officer**

Analytical Approach

Data Understanding

Understanding the structure and characteristics of data, and detect potential early problems.

Data Preprocessing

Cleaning data with imputation techniques (KNN or Iterative), handling outliers, and encoding and scaling variables.

Modeling

Using and experimenting with various machine learning algorithms until we find the one that best suits our customer churn predictions.

Exploratory Data Analysis (EDA)

Conduct descriptive analysis and data visualization to identify relationships between variables and patterns that may emerge.

Feature Engineering

Perform binning, create new relevant features, and select features that have a significant impact on the prediction target.

Evaluation

Evaluate model performance using metrics that are appropriate to the objectives and business problems being solved.

Evaluation Metrics

Company is focused on **capturing** as many customers as possible who are truly at **risk of churn** (FalseNegative minimization) while still maintaining retention cost efficiency (FalsePositive control), the most appropriate metric is the **F2-Score**.

$$\text{F2 - Score} = \frac{(1 + 2^2) \times \text{Precision} \times \text{Recall}}{(2^2) \times \text{Precision} + \text{Recall}}$$

FN = False Negatives = Model wrongly predict that the churned customer as not churned

FP = False Positives = Model wrongly predict that the not churned customer as churned

F2 Score

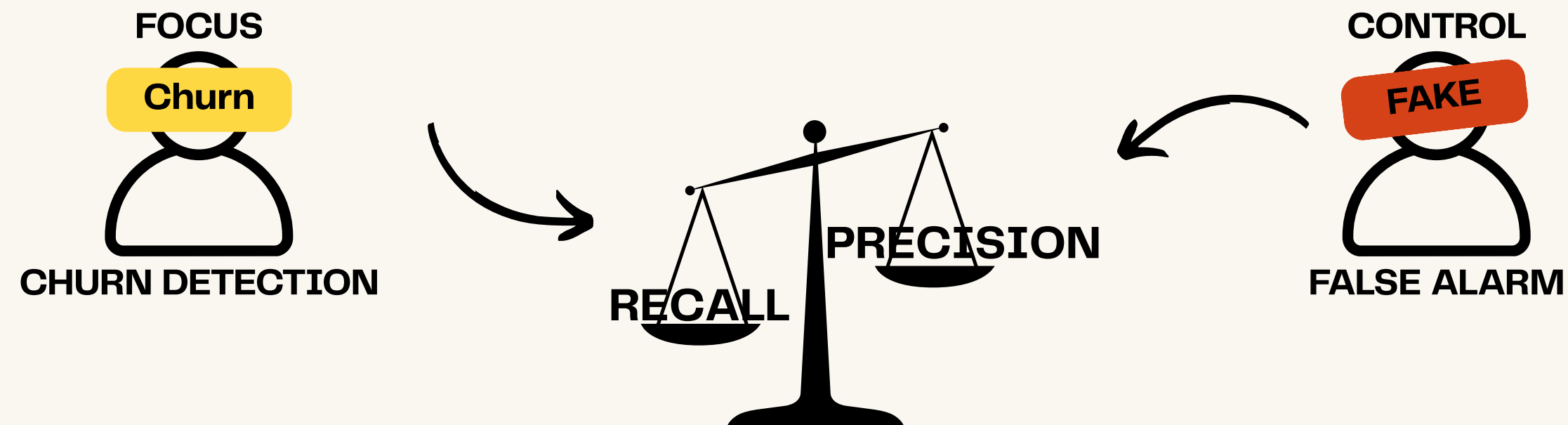
F2-Score gives **greater weight** to **Recall** (detecting as many churned customers as possible) than Precision, but still **maintains** a **balance** between the two.

Evaluation Metrics : F2

Objective

Capturing as many customers as possible who are truly at **risk of churn**, while **maintaining retention** cost efficiency.

Why F2 ?



F2 Score

gives **greater weight** to **Recall** (detecting as many churned customers as possible) than Precision, but still **maintains a balance** between the two.



Exploratory Data Analysis

+



Data Understanding

15 Numeric Columns

5 Category Columns

30 Anomalies

1856 Missing Values

Customer Profile

CustomerID

Unique ID of each customer.

Gender

Customer gender.

MaritalStatus

Customer's marital status.

Operational

WarehouseToHome

Distance between warehouse and customer house (km).

NumberOfAddress

Number of addresses registered to the customer.

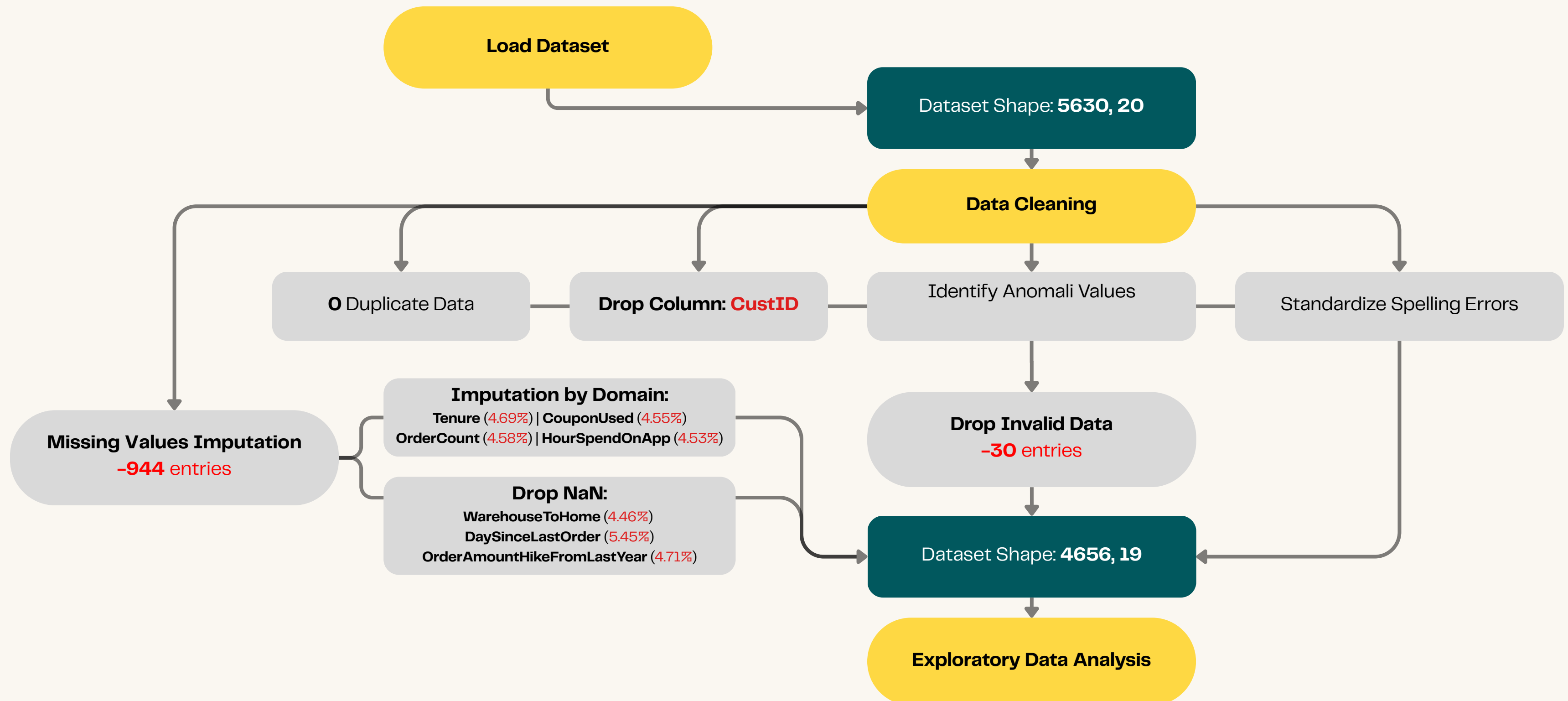
CityTier

Tier of the city where the customer lives.

Data Understanding

Transactional		Customer Experience	
Tenure	Tenure of customer in organization.	Churn	Status whether the customer has churned or not.
OrderCount	Total number of orders has been places in last month.	PreferredLoginDevice	The primary login device the customer uses.
OrderAmountHike FromLastYear	Percentage increase in the number of orders compared to the previous year.	HourSpendOnApp	The number of hours customers spend on the app/website.
CouponUsed	Total number of coupon has been used in last month.	SatisfactionScore	Customer satisfaction score for the service.
CashbackAmount	Average cashback in last month (\$).	Complain	Have customers filed complaints in the last 1 month.
DaySinceLastOrder	Day since last order by customer.		
PreferedOrderCat	Most popular order categories in the last month by a customer.		
PreferredPaymentMode	The payment method most frequently used by customers.		

Data Preparation Journey



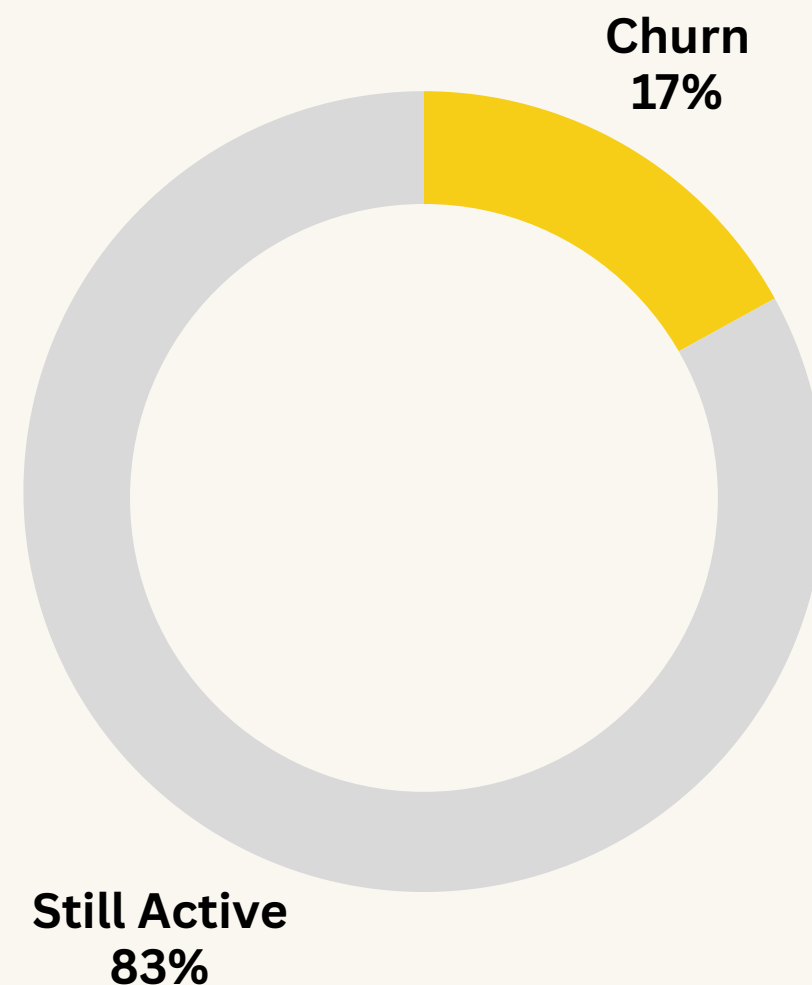
Churn Landscape

Churn rate **greater than *5-10%** is a signal of **potential issues** that could **affecting business health**.

4,7K
Customers

17%
Churn Rate

Alpha Customer Churn Distribution



1%_{pp} × 4,7k cust × \$60 = **\$2,8K**

↓ 5%_{pp} × 4,7k cust × \$60 = **\$14K** 

Key Findings:

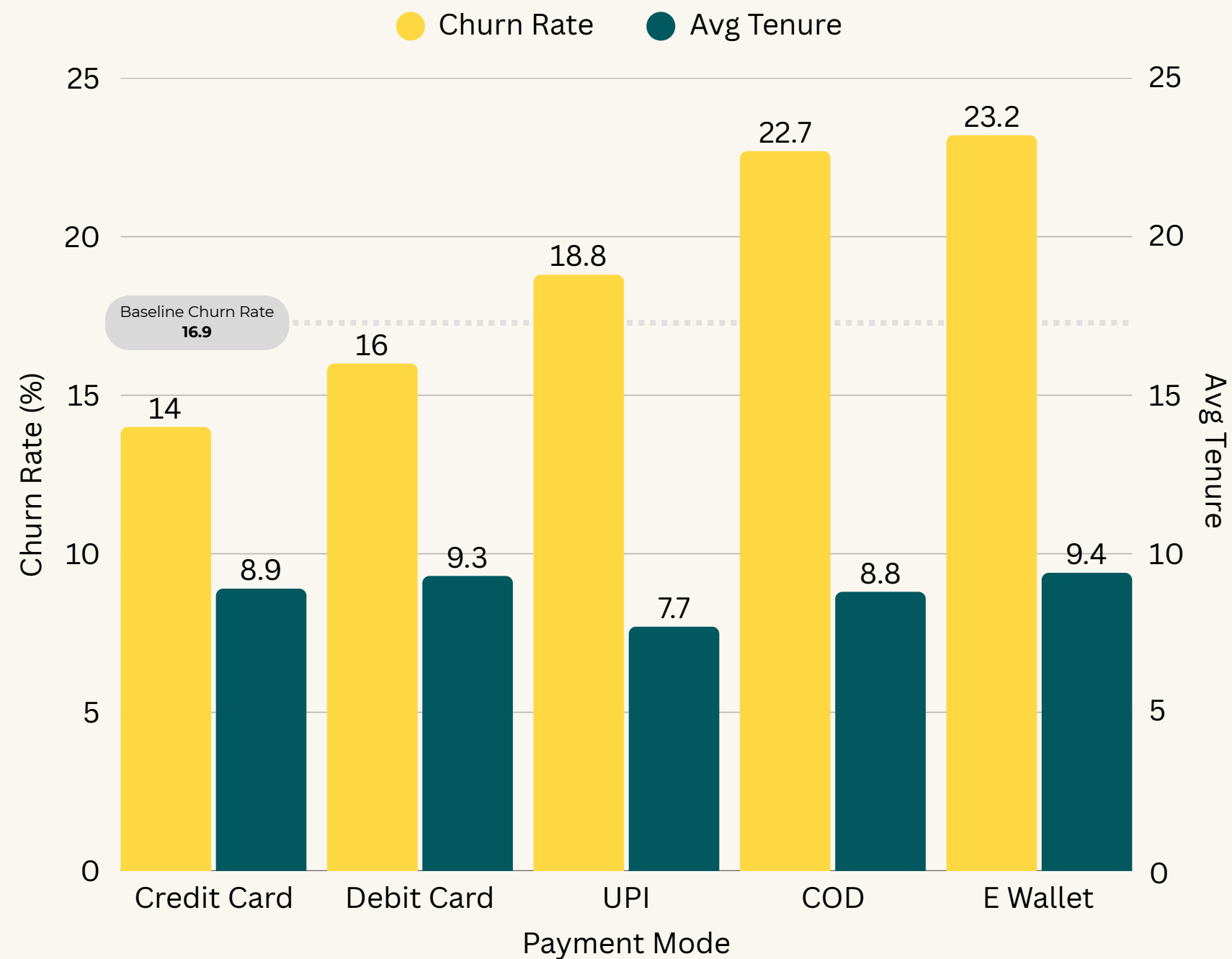
17% customer churn, **higher than** the company's target (**≤12%**).

5-point decrease could **save** approximately **\$14K** in **new customer acquisition costs**.

*Source: Ben Salomon, Yotpo, "Churn Rate: Calculation, Importance & Reduction Strategies", 2025

The Hidden Cost of Instant Payments

Churn Rate (%) by Payment Mode



Key Findings:

Instant payment methods (**E Wallet, COD & UPI**) show the **highest churn**.

E-Wallet users churn at **23.2%** but **stay longer** on average.

Card users are the **most stable**, with churn around **14–16%**.

Payment Mode Improvements:



Shift customers to **digital payments**.



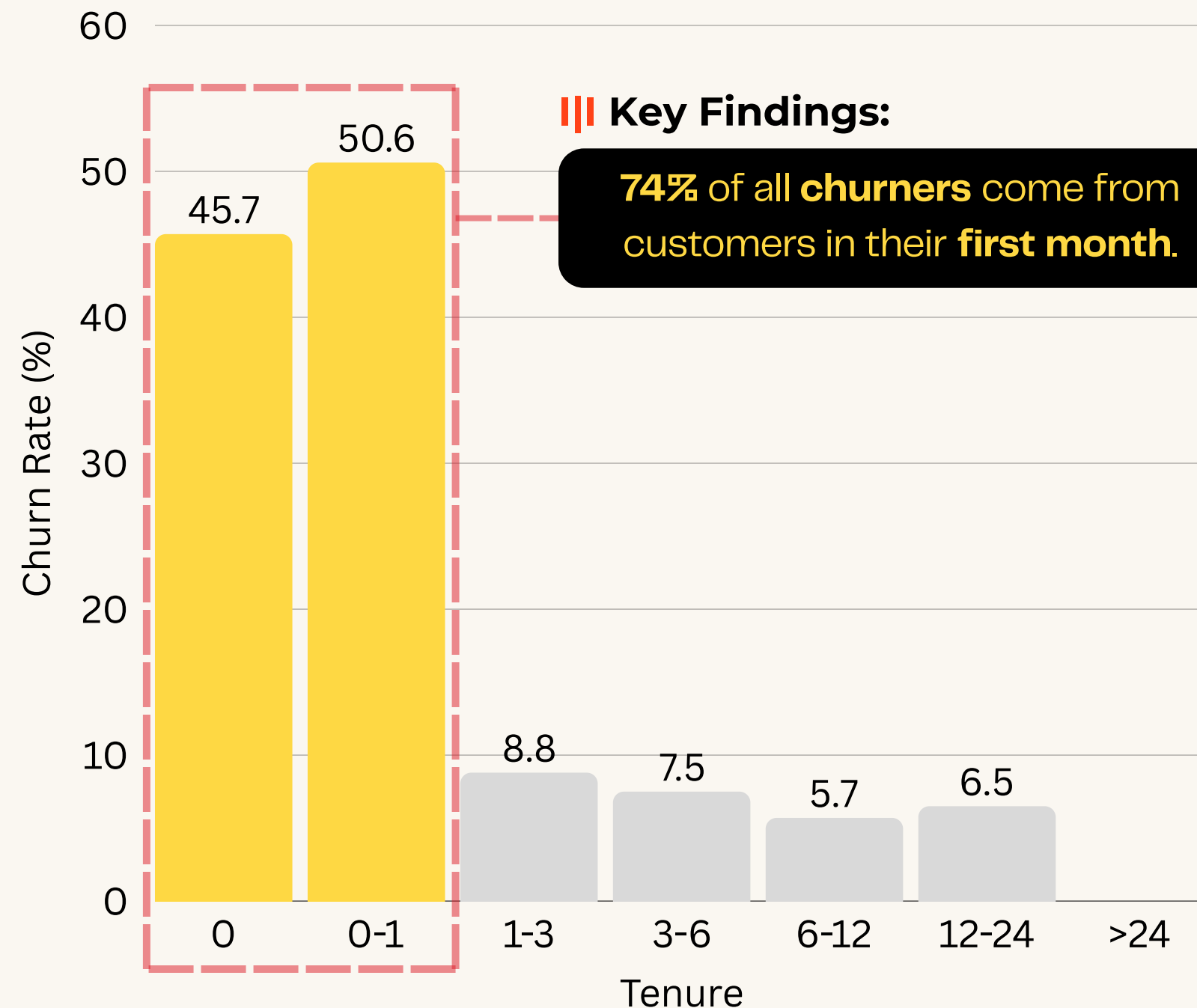
Reward card users on their **2nd** or **3rd** purchases.



Flag and **follow up COD users** with **inactivity >7 days** for **targeted** win-back offers.

The First 30 Days Decide Everything

Average Churn Rate (%) by Tenure



“We lose them before they even start.”

Retention Starts on Day One:



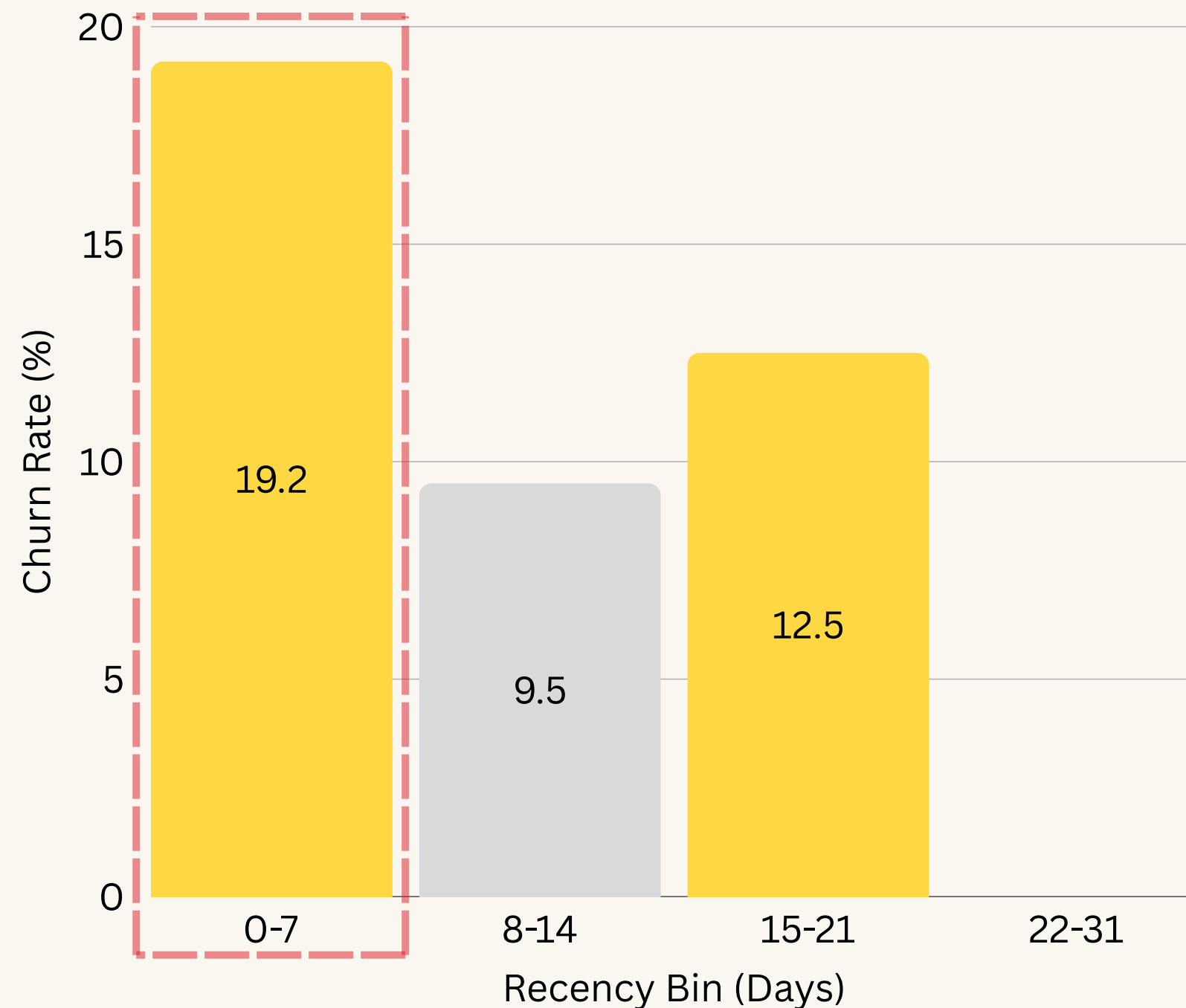
Prioritize **“First-90-Days Retention Program”** powered by **Machine Learning** predictions.



Triggered Retention for **New Customers (0–30 Days)**.

The First Week is When We Lose Them

Average Churn Rate (%) by Recency



Key Findings:

19% churn **happens** within the **first 7 days of inactivity**.

Customers who **stay after one week** are much more likely to **remain**.

Indicates an onboarding or **first-experience problem**, not long-term disengagement.

Onboarding Strategies:



Trigger retention in the first week.



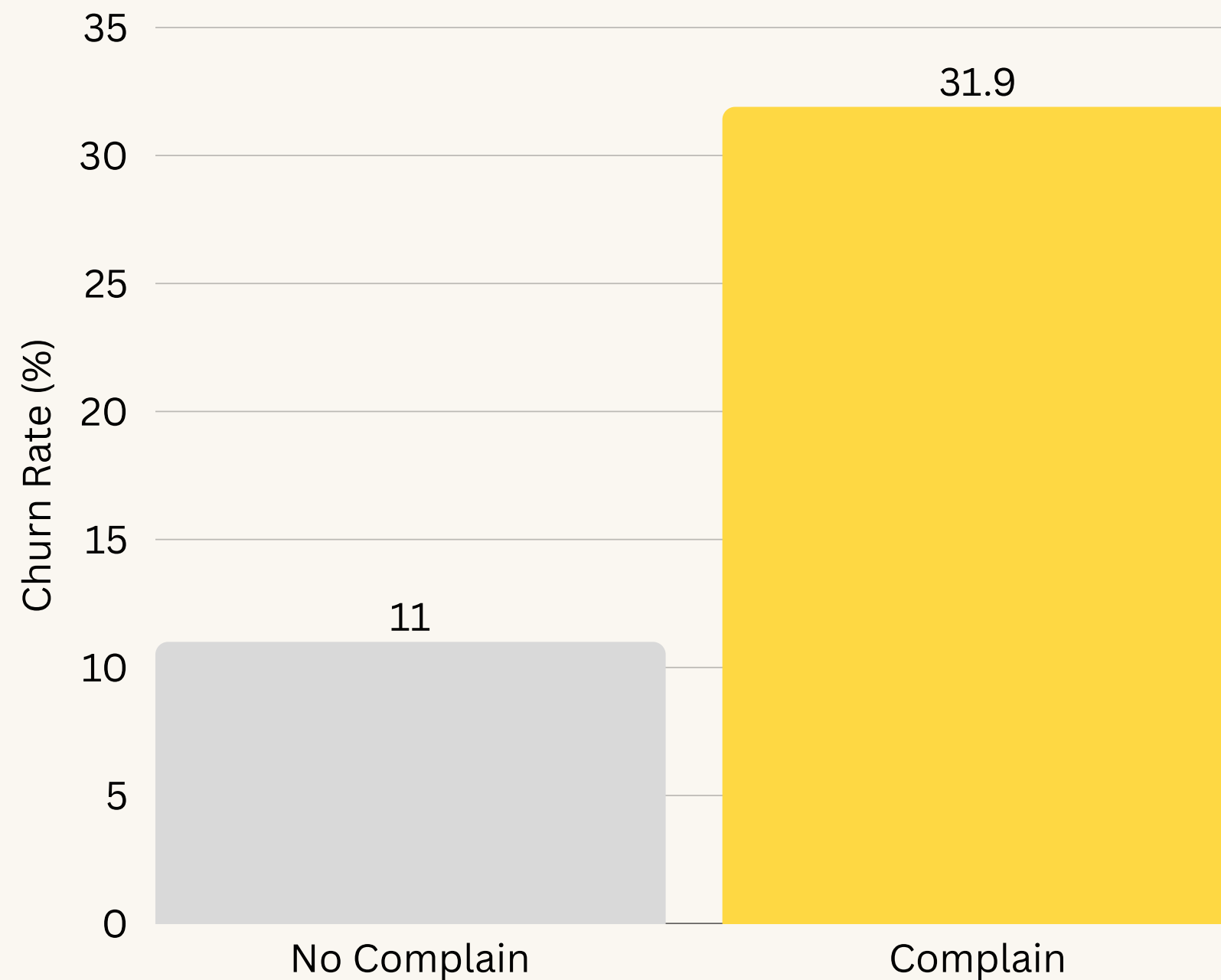
Measure **“Activation within 7 Days”** as an **early KPI** for **new users**.



Automate win-back offers for **7-14-day inactive users**.

Complaints Are Our Earliest Warning

Churn Rate (%) by Complaint Status



Key Findings:

Customers who **complained** are **2.9× more likely** to **churn**.

53.6% of all **churners** had made **at least one** complaint.

CS Recovery Actions:



Implement **“Immediate Recovery”** Workflow.



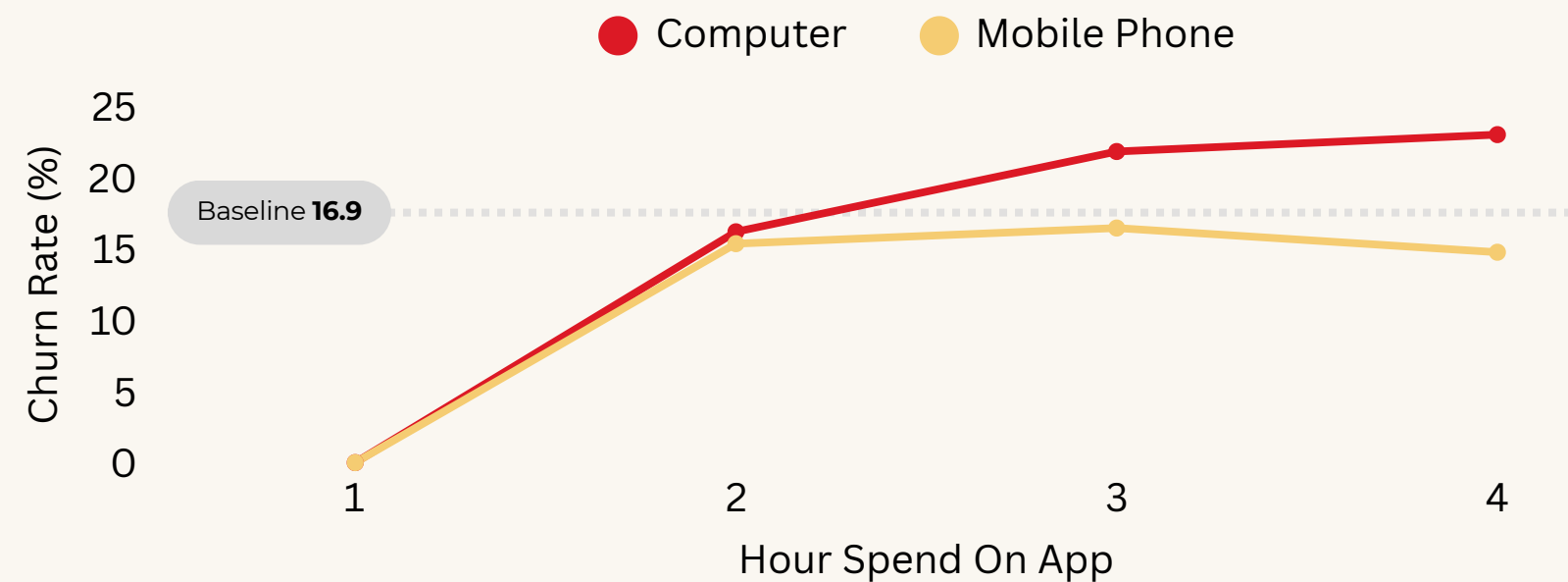
Prioritize Complainers in Retention Campaigns.



Track **“Complaint Resolution Time”** as new **retention KPI**.

High Engagement, Hidden Frustration

Churn Rate (%) by Device & Hour Spend On App



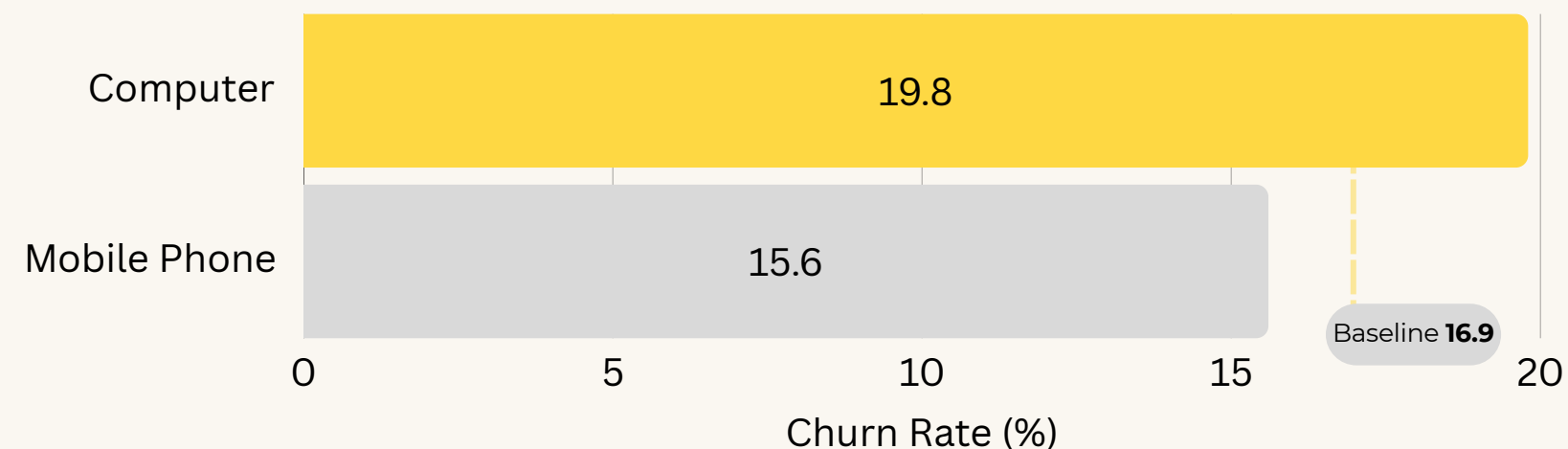
Key Findings:

Desktop users show the **highest** churn.

Heavy app users (>3h/day) **churn more**.

High activity ≠ satisfaction, indicating friction, not loyalty.

Churn Rate (%) by Preferred Login Device



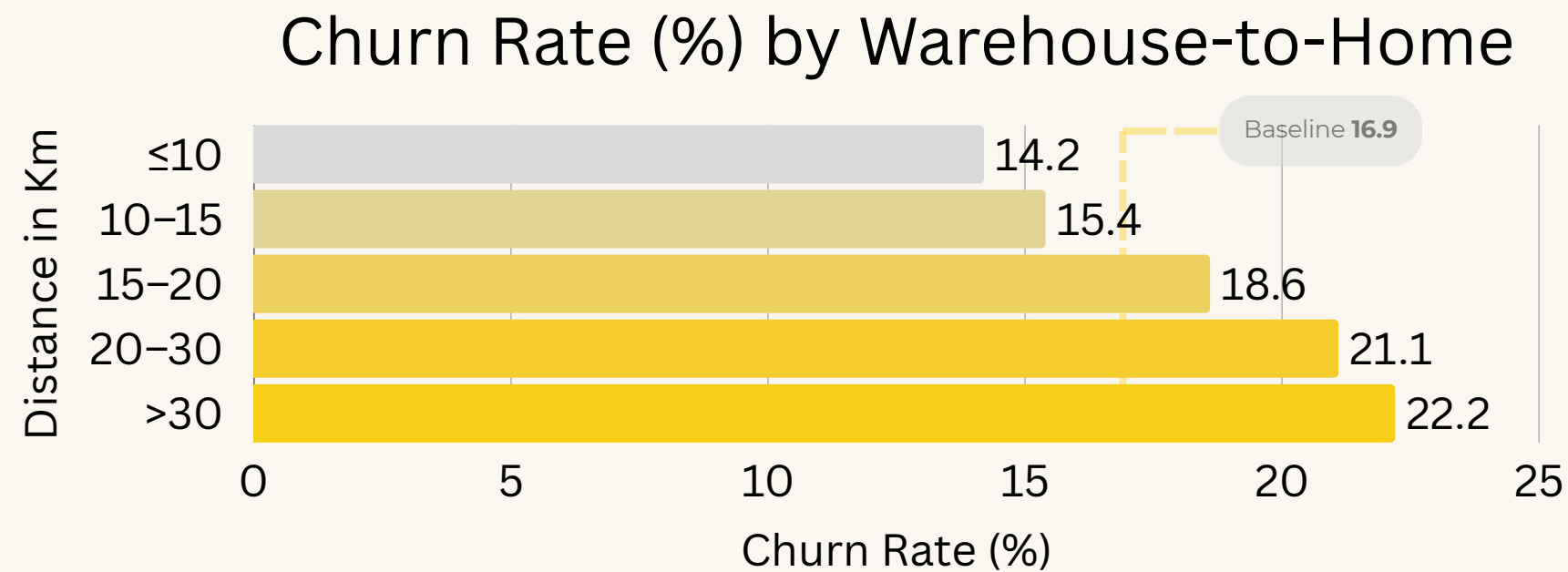
Engagement & CX Actions:

Partner with IT to **align web & app UX**, and **run A/B tests** to **identify friction points** between platforms.

Track long sessions with **low conversion**.

Retain heavy users via **personalized offers**.

Logistics Quality Matters

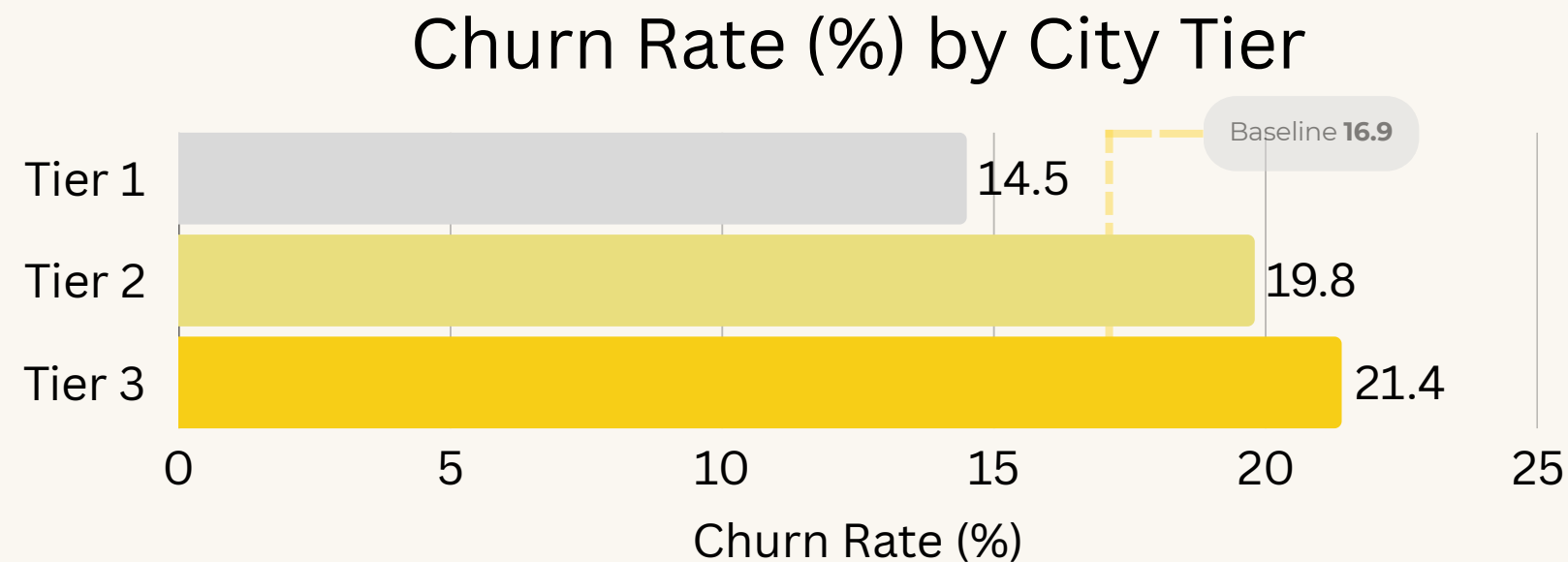


Key Findings:

Churn **rises steadily** with **delivery distance**.

Customers **beyond 30 km** churn at **22%**, ~5 pp above average.

Tier-3 cities **mirror** the same **distance-driven risk**.



Fulfillment Actions to Cut Churn:



Align **delivery ETA** with **actual fulfillment capacity**.



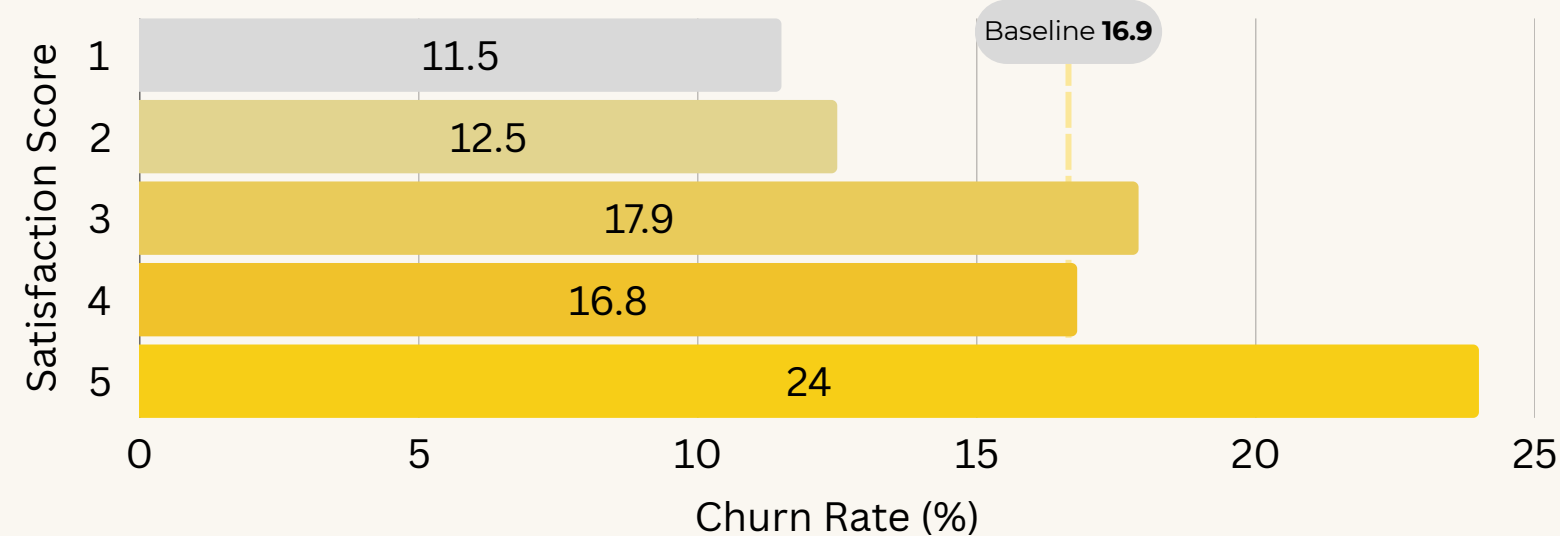
Improve **delivery speed** in **Tier-2** and **Tier-3** areas.



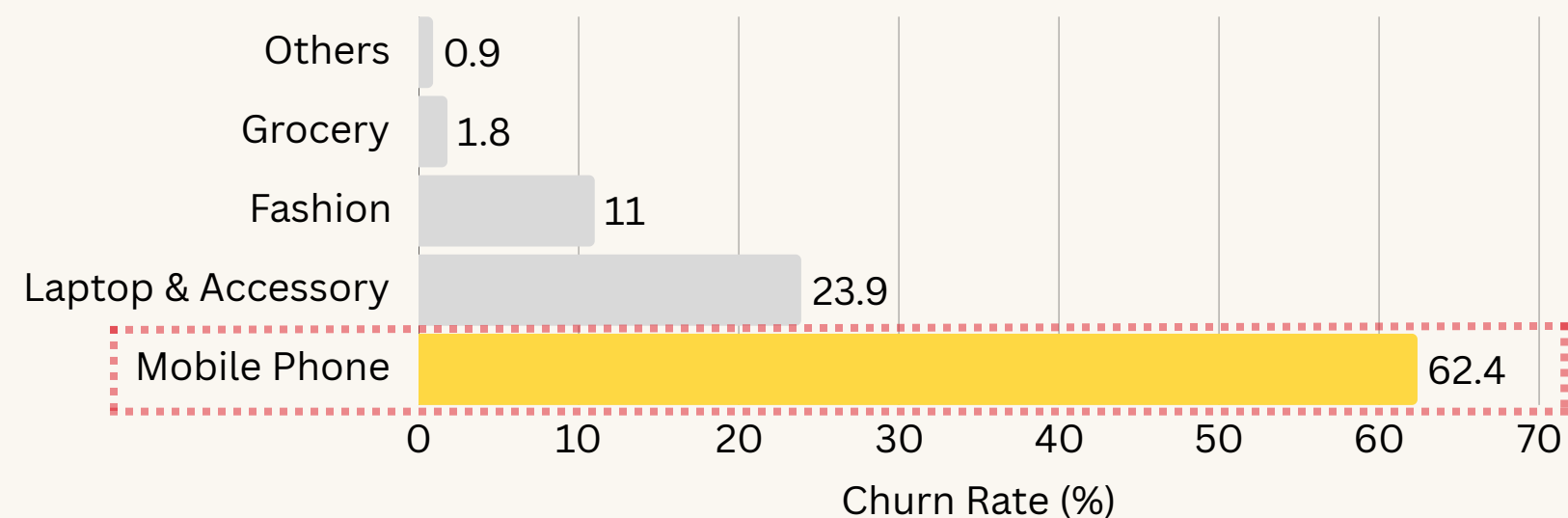
Closely **monitor long-distance orders** to prevent churn.

When Satisfaction Doesn't Mean Loyalty

Churn Rate (%) by Satisfaction Score



Share of 'Happy Churners' (CSAT=5) by Preferred Order Category



Key Findings:

Satisfaction doesn't guarantee loyalty.

True **churn drivers** are **operational**, not **sentiment**.

High-value categories indicating **"happy churners"**.

Customer Insight Actions:



Track **satisfaction** and **retention** separately.



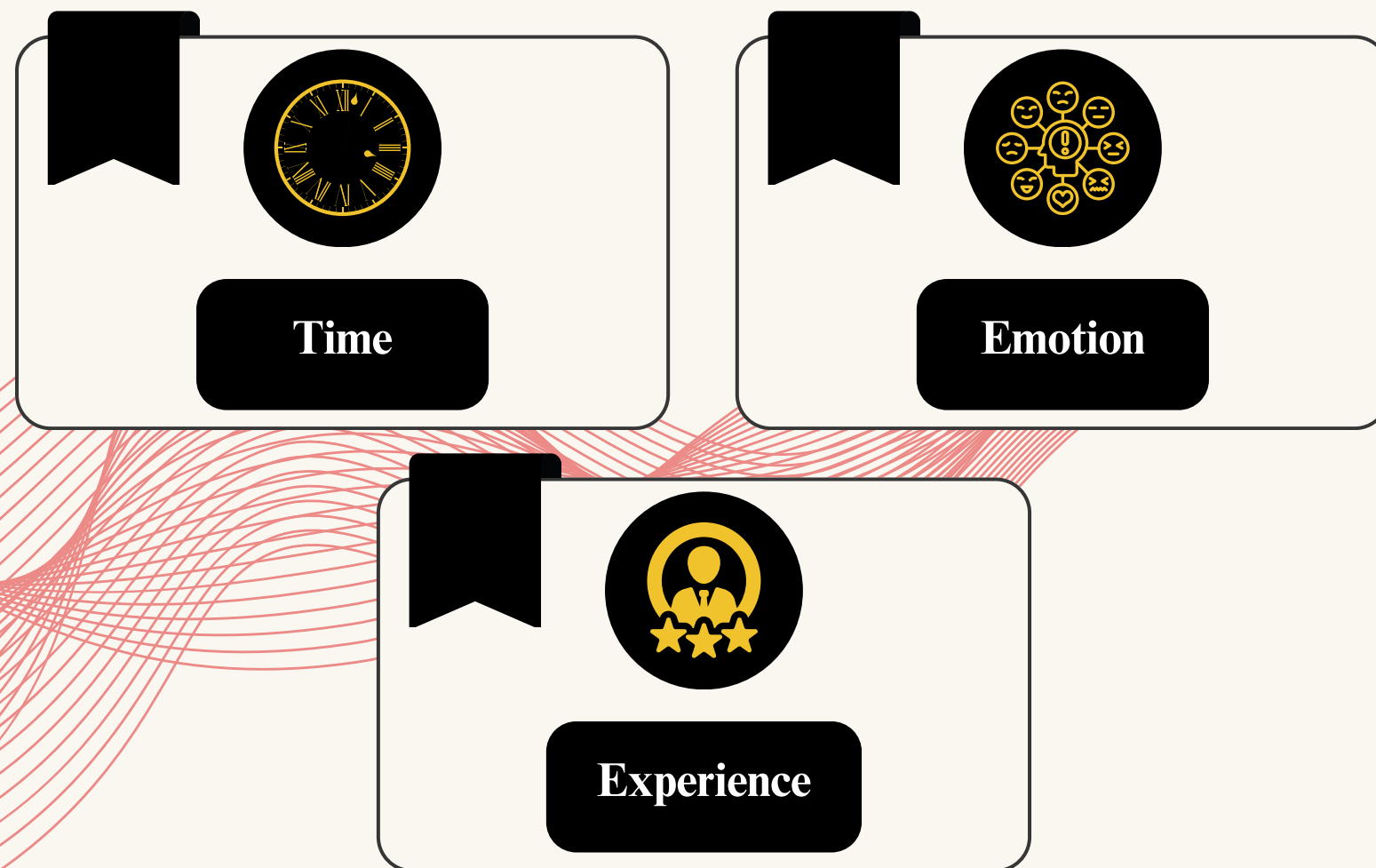
Add **delivery** & **refund** experience in **satisfaction tracking**.



Re-engage "happy churners" with **win-back offers**.

We Don't Lose Customers Because of Who They Are
We Lose Them Because of How They Behave

Where, When, and Why Customers Walk Away:





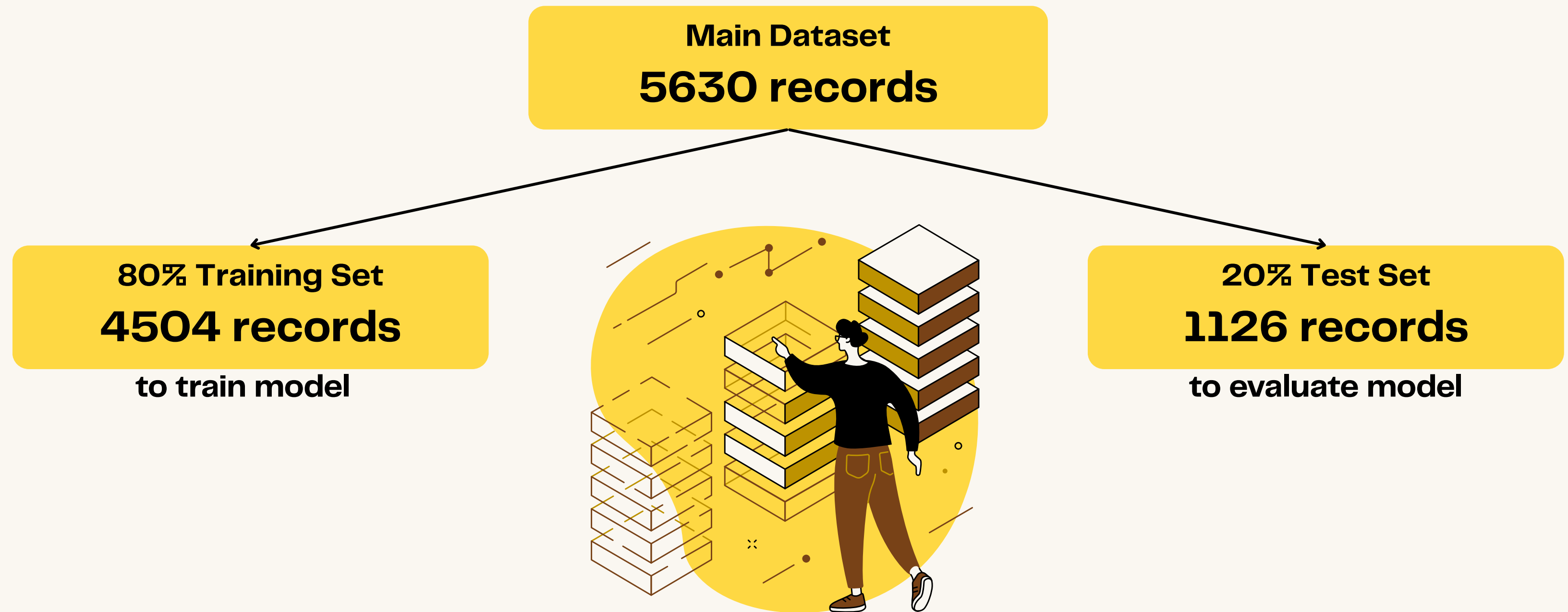
Modeling & Interpretation

+



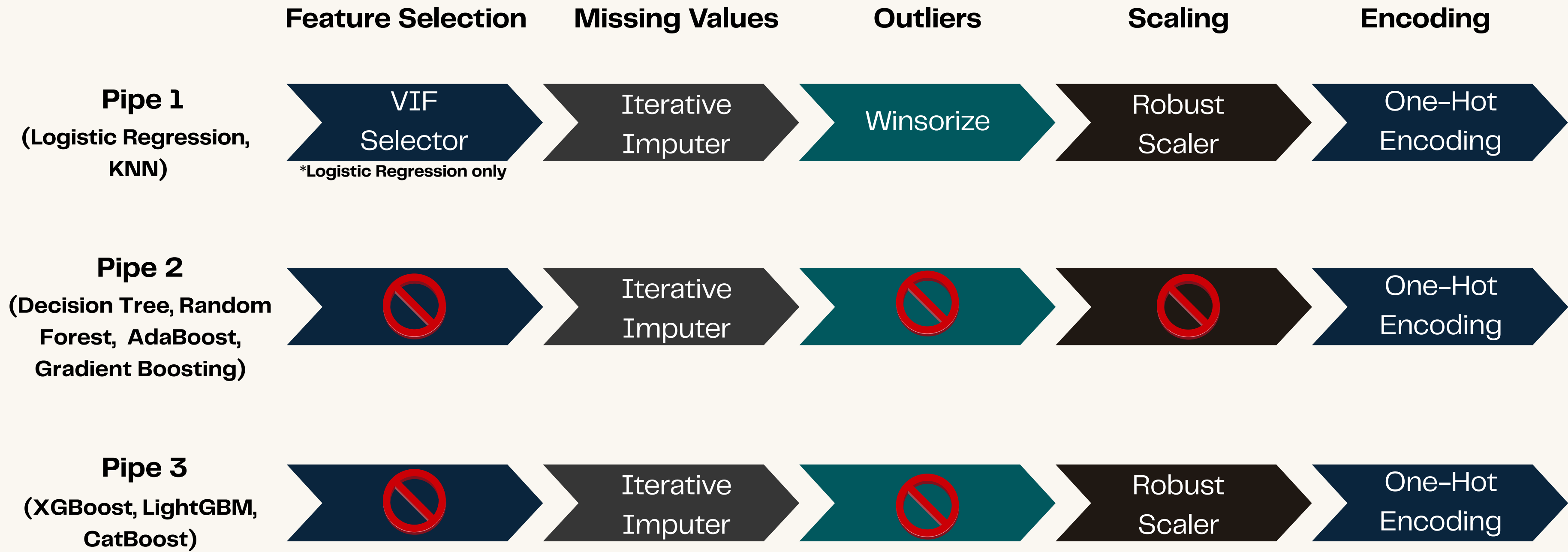
Data Splitting

Construct Training and Testing Data



Modeling Strategy

Treat the models as it should be



Base Model Evaluation

	Model	Avg CV F2 Score	Std CV F2 Score
1	XGBoost	0.8531	0.0360 
2	LightGBM	0.8273	0.0348
3	Decision Tree	0.8011	0.0163
4	Random Forest	0.8007	0.0358
+5 records...			

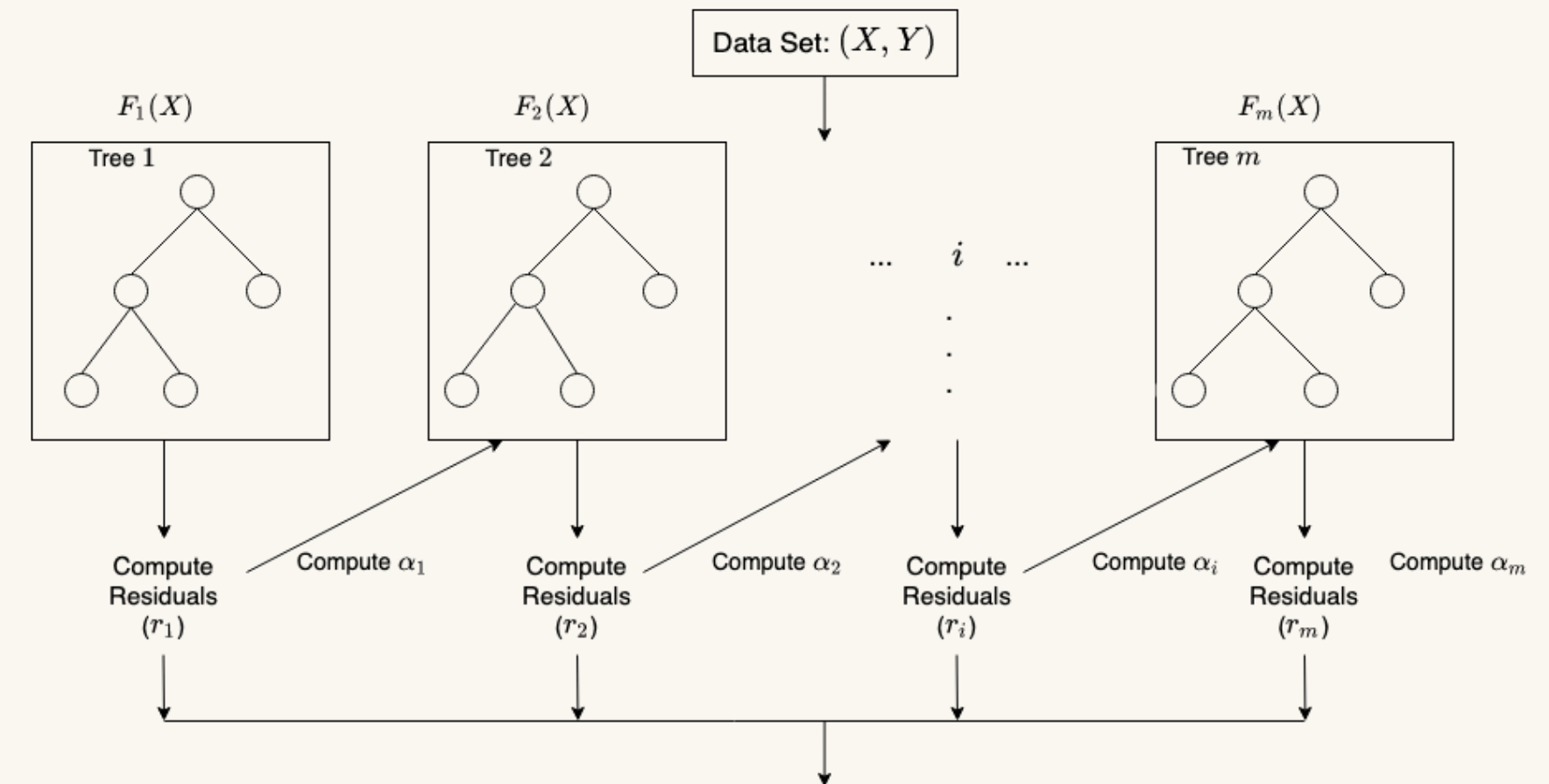
Best Model: XGBoost[👑]

XGBoost is an ensemble learning method that combines multiple decision trees sequentially to minimize prediction errors. It uses gradient descent optimization to minimize the overall loss function, improving prediction accuracy with each iteration. The final model is a weighted combination of all trees, producing strong predictive performance.

How it works:

1. The first tree makes initial predictions → calculate the errors.
2. The next trees are built to correct those errors.
3. This process continues iteratively to improve performance.
4. The final prediction = weighted sum of all trees.

$$\hat{y} = \sum_{k=1}^K f_k(x)$$



$$F_m(X) = F_{m-1}(X) + \alpha_m h_m(X, r_{m-1}),$$

where α_i , and r_i are the regularization parameters and residuals computed with the i^{th} tree respectively, and h_i is a function that is trained to predict residuals, r_i using X for the i^{th} tree. To compute α_i we use the residuals

computed, r_i and compute the following: $\arg \min_{\alpha} = \sum_{i=1}^m L(Y_i, F_{i-1}(X_i) + \alpha h_i(X_i, r_{i-1}))$ where $L(Y, F(X))$ is a differentiable loss function.

Source : AWS

Resampling Experiment

with XGBoost model

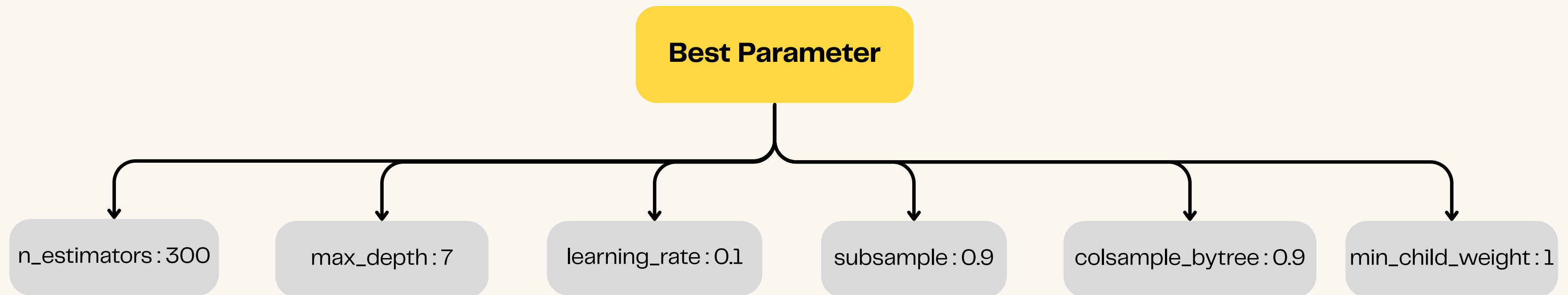
	Method	Avg CV F2 Score	Std CV F2 Score
1	class_weight = 'balanced'	0.8816	0.0165 
2	Random Over Sampler	0.8795	0.0190
3	none	0.8619	0.0265
4	SMOTE	0.8599	0.0276
5	Random Under Sampler	0.8342	0.0217

XGBoost Hyperparameter Tuning

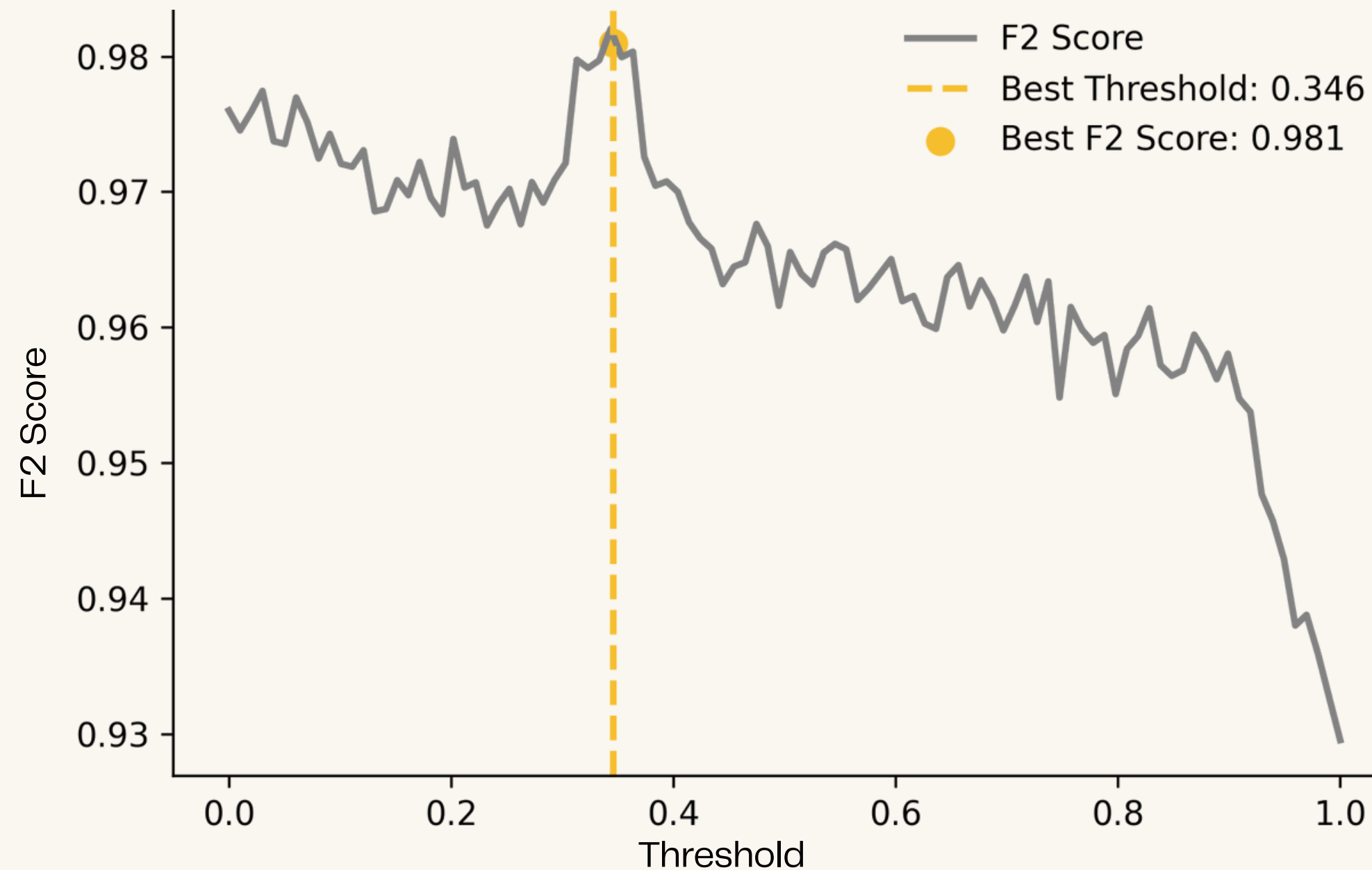
Parameter	Description	Range
n_estimators	The total number of decision trees to be built by XGBoost.	[200, 300, 500]
max_depth	The maximum depth of each decision tree.	[3, 5, 7, 9]
learning_rate	Controls how much each new tree contributes to the overall model.	[0.01, 0.05, 0.1, 0.2]
subsample	The fraction of training data used to build each tree.	[0.8, 0.9, 1.0]
colsample_bytree	The fraction of features (columns) randomly sampled for each tree.	[0.8, 0.9, 1.0]
min_child_weight	The minimum sum of instance weights (Hessian) required in a child node.	[1, 3, 5]

Final Performance

Avg CV F2-Score	Std CV F2 Score	Test F2 Score
0.8894	0.0216	0.9759
+0.078	+0.051	



Threshold Optimization



The optimal threshold is 0.346

F2 score can increase up to 0.52%

The optimal probability threshold achieves a balance between precision and recall.

When **the threshold is set too low** (around 0.1–0.2), nearly all customers are **predicted as “Churn”**, which causes **precision to drop significantly (more True Positive)**.

On the other hand, if **the threshold is set too high**, most customers are **classified as “Not Churn”**, leading to a considerable **decrease in recall (more True Negative)**.

Features Evaluation



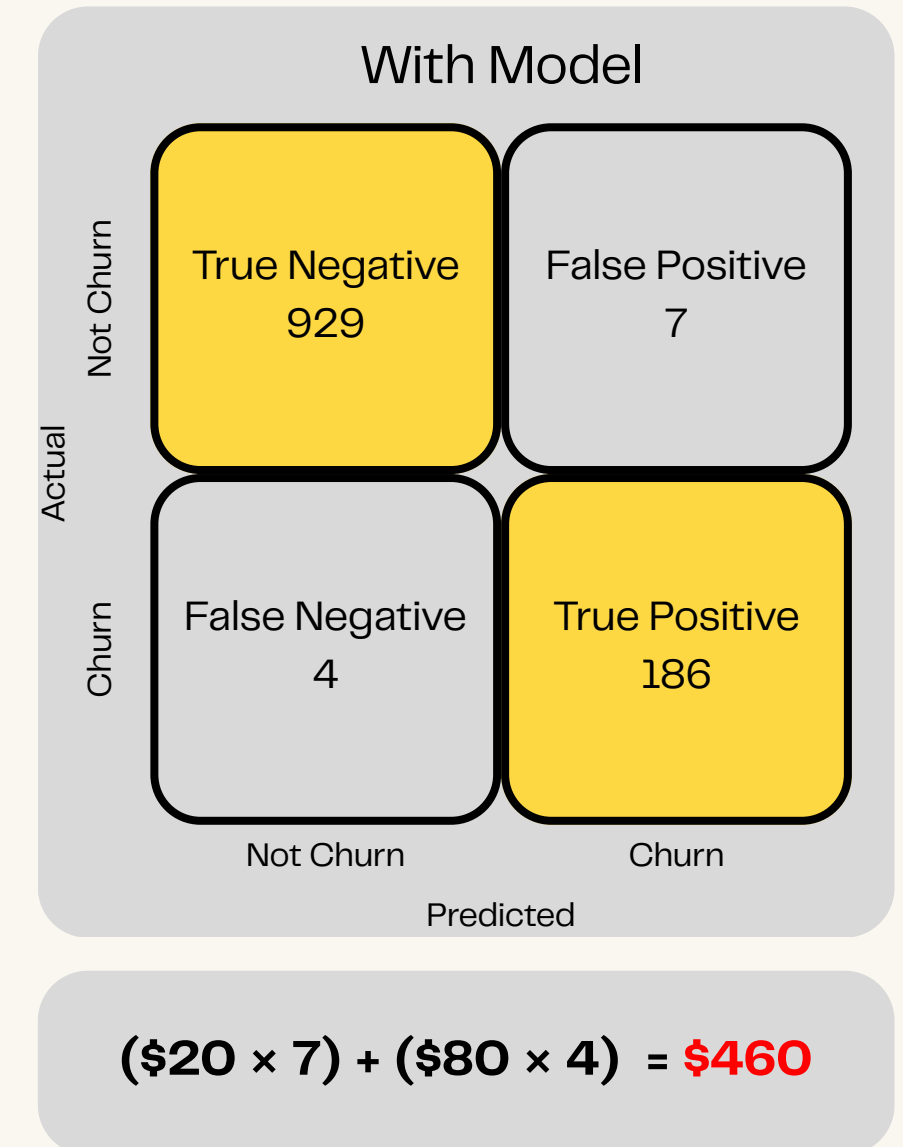
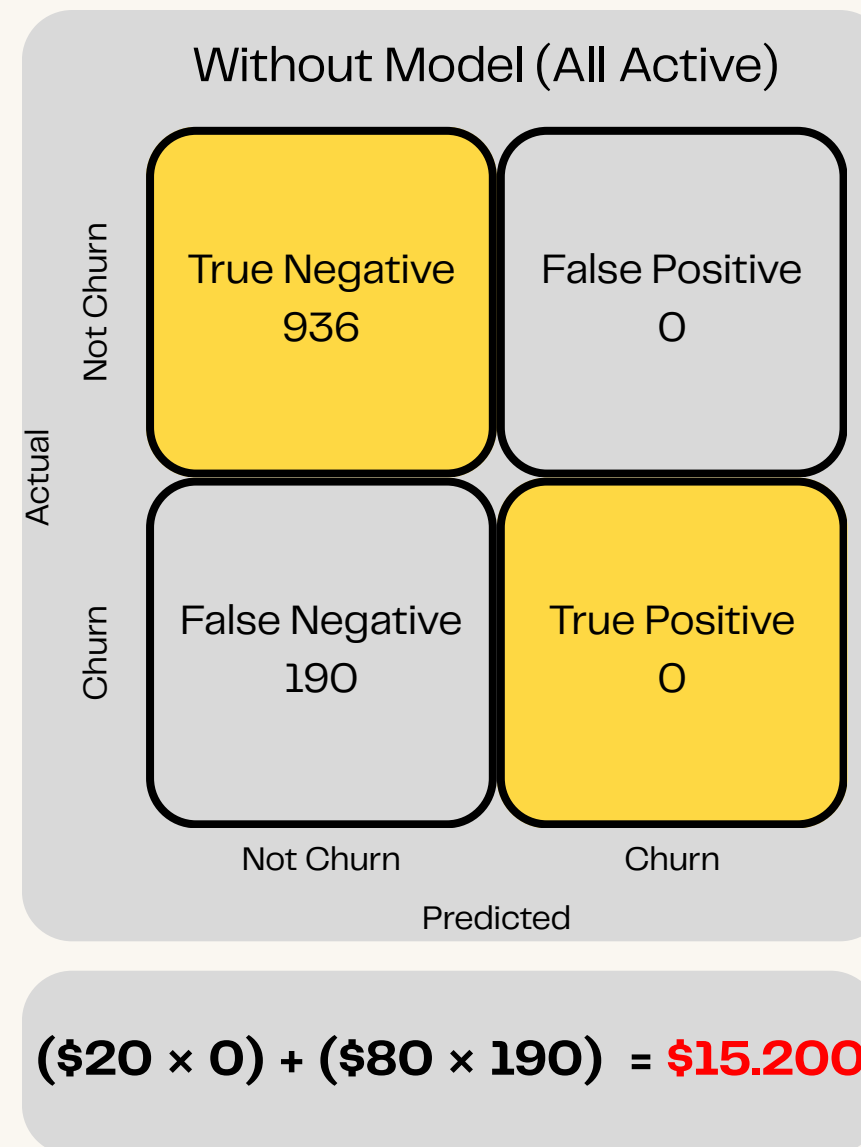
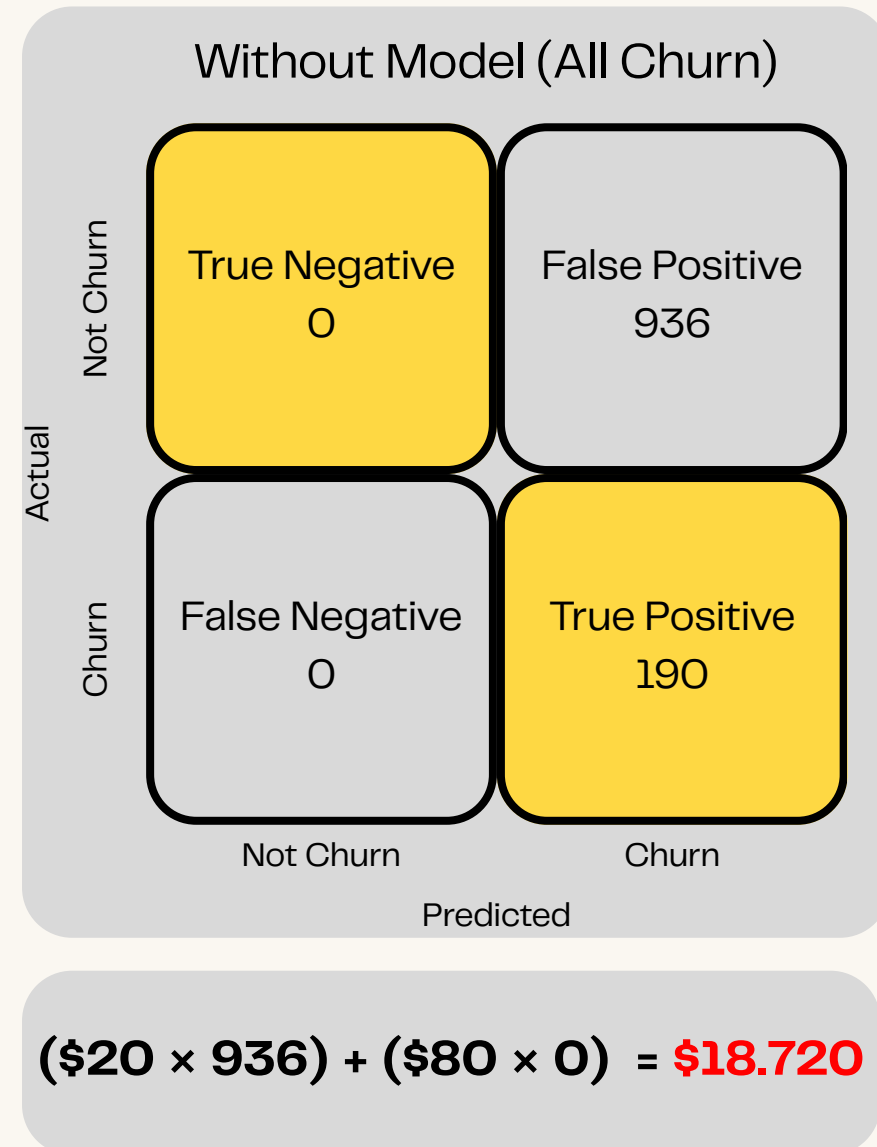
Features that Increase Churn Risk

- **Short tenure (+5)** — new customers are more likely to churn
- **High complaint count (+4)** — indicates dissatisfaction
- **More addresses (+4)** — reflects lower stability or loyalty
- **Low cashback amount (+4)** — feel less benefited
- **Long warehouse distance (+3)** — longer delivery time reduces satisfaction
- **Long time since last order (+2.5)** — shows declining engagement
- **Single customers & higher city tier (+1.5)** — slightly higher churn tendency

Features that Reduce Churn Risk

- **High tenure (-7)** — indicates loyalty
- **Less device registered (-3)** — indicates stronger engagement
- **High cashback amount (-3)** — incentives encourage retention
- **Close from warehouse (-3)** — shorter delivery time
- **Growth in annual order amount (-2.5)** — spending increase signals loyalty

Business Impact

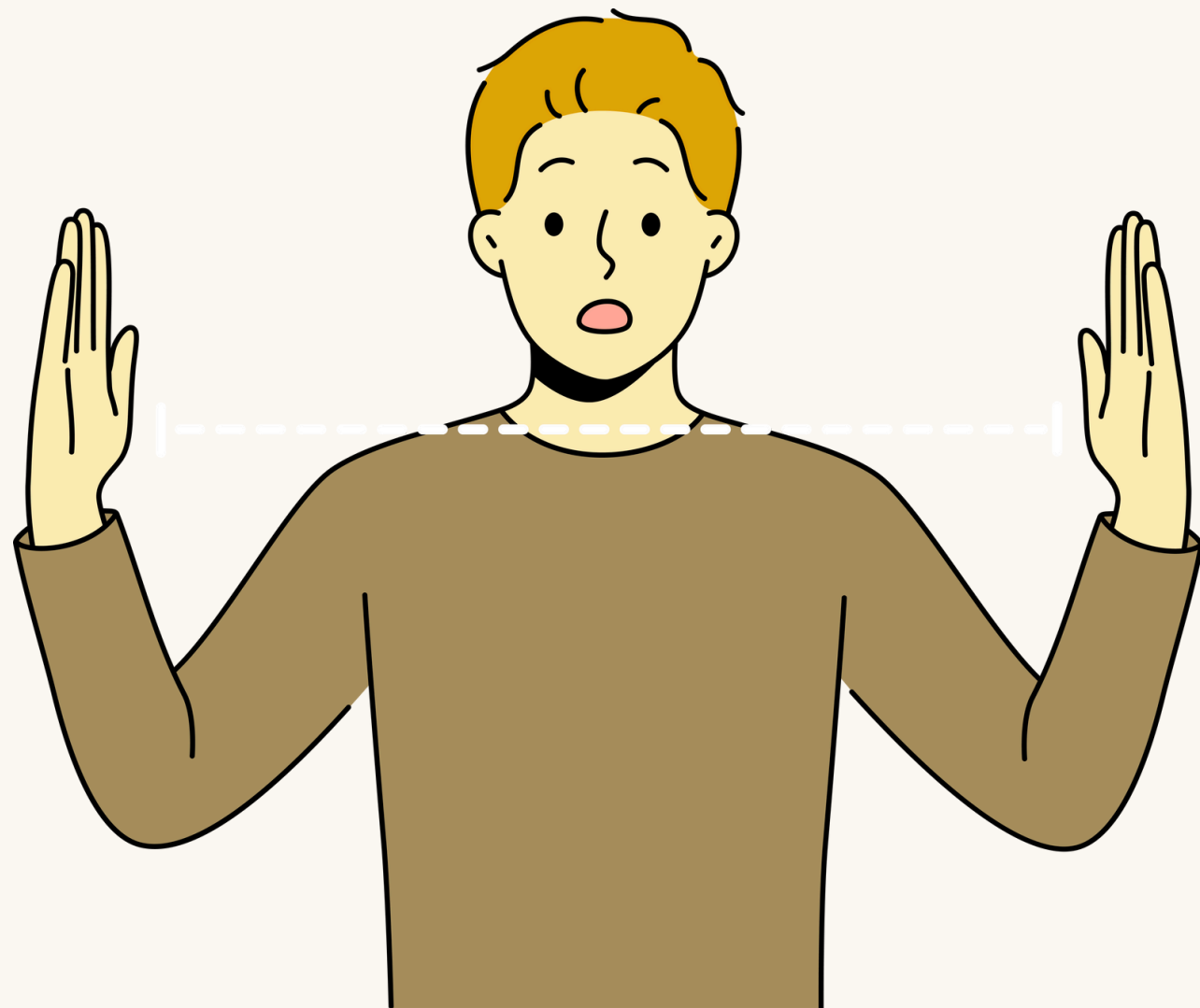


Assumption :

False Positive	False Negative
\$20	\$80

With the model, the business loss drops drastically to only \$460, compared to \$18,7K (40×) and \$15,2K (33×) without the model — showing a major improvement in cost efficiency and customer retention targeting.

Model Limitations



- The definition of churn is a bit ambiguous.
- Less suitable when precision or recall alone needs to be emphasized.
- No seasonal effect consideration.
- Missing monetary feature for more complexity.
- Do not suitable with linear model.



Conclusion & Recommendations

+



Churn Summary



New or short-tenure users.

Inactive for long periods.

Frequent complaints.

Delivery delays.

Rely on short-term promos.

Use Cash on Delivery method.

Computer access.

Low cashback gained.

Model Summary



The developed XGBoost machine learning model notching achievements such as :

- Achieved very high F2 Score (+97%).
- Identified the influence of features.
- Reduce the financial loss up to 40x.

Business Recommendations

Early Retention Campaign (First 90 Days)

Provide personalized onboarding journeys (tutorials, welcome offers, guided app tours) to help new or short-tenure users quickly find value in the platform.

Offer limited-time discounts or early rewards to encourage repeated orders and build engagement habits within the first 90 days.

Optimize Retention Budget Allocation

Use model predictions to **prioritize high-risk and low-engagement customers** (inactive, COD users, promo-dependent). Allocate incentives strategically with **A/B-tested frequency caps** to avoid overspending while maintaining optimal Return on Investment (ROI) and maximizing reactivation impact.

Cross-Channel Re-Engagement & Experience Optimization

Identify inactive or low-engagement users through inactivity tracking and trigger personalized reactivation campaigns via multiple channels (email, push, WhatsApp). Encourage app adoption for desktop or multi-device users by offering exclusive app-only rewards and smoother UX to reduce friction and strengthen long-term engagement.

Enable Complaint Recovery

Identify frequent complaints and implement rapid recovery workflows at the operational level. Provide personalized compensation (e.g., vouchers, delivery fee waivers) to rebuild trust and follow up to ensure satisfaction, turning negative experiences into loyalty opportunities.

Loyalty Program & Engagement Shift

Create tiered loyalty programs to reduce reliance on short-term promos. Reward consistent activity and digital payments with exclusive perks (e.g., cashback bonuses, early access, free shipping), fostering long-term emotional attachment and stable retention.



Model Recommendations

Feature Engineering Improvements

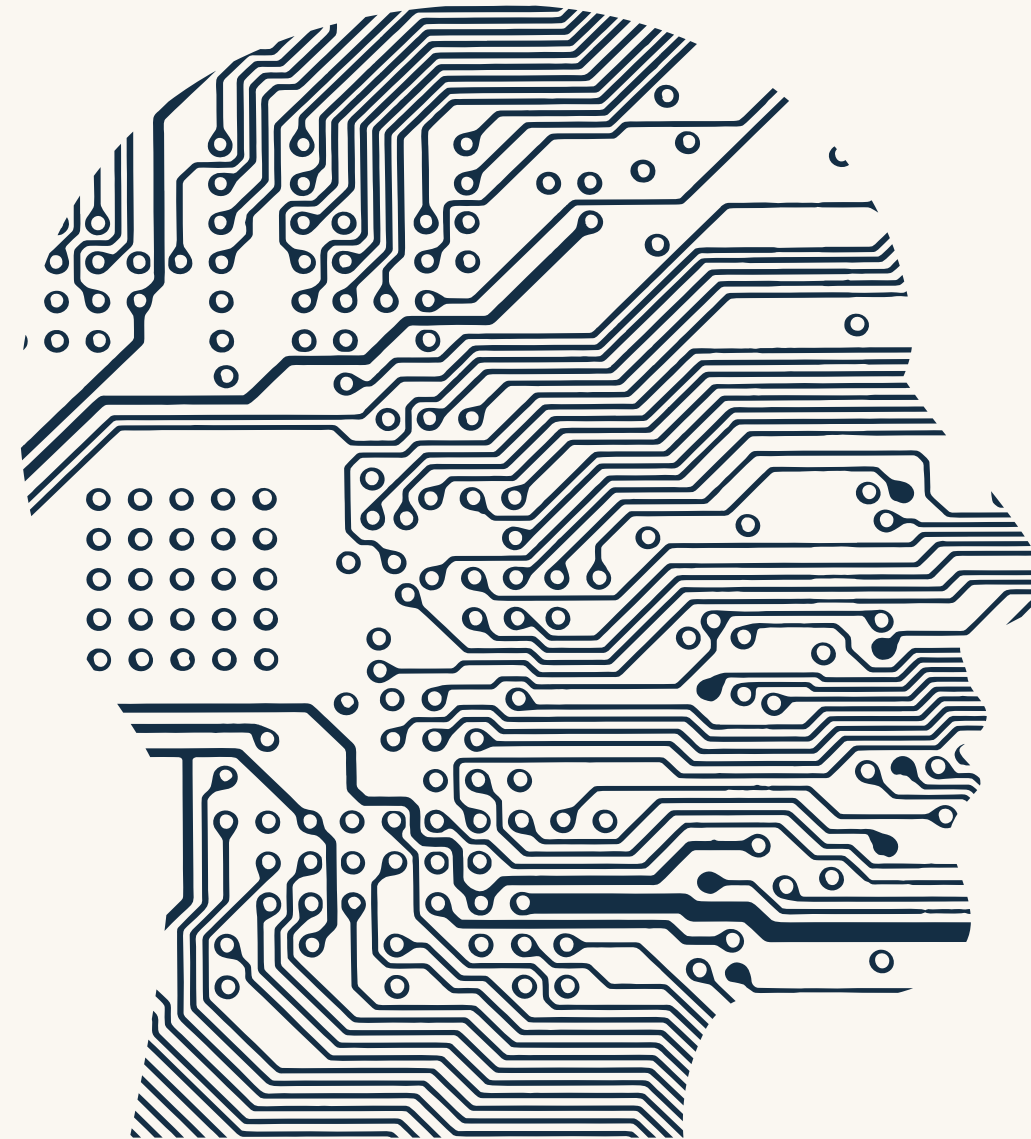
Add **monetary-related features** (e.g., total spending, purchase value) to capture **Customer Lifetime Value (CLV)**.

Optimize Hyperparameters

Enable **Grid Search** with more type or range of **hyperparameter** for the best performance.

More Resampling Experiment

Try to use **another resampling method** (e.g., Near Miss, ADASYN) for more experiment.



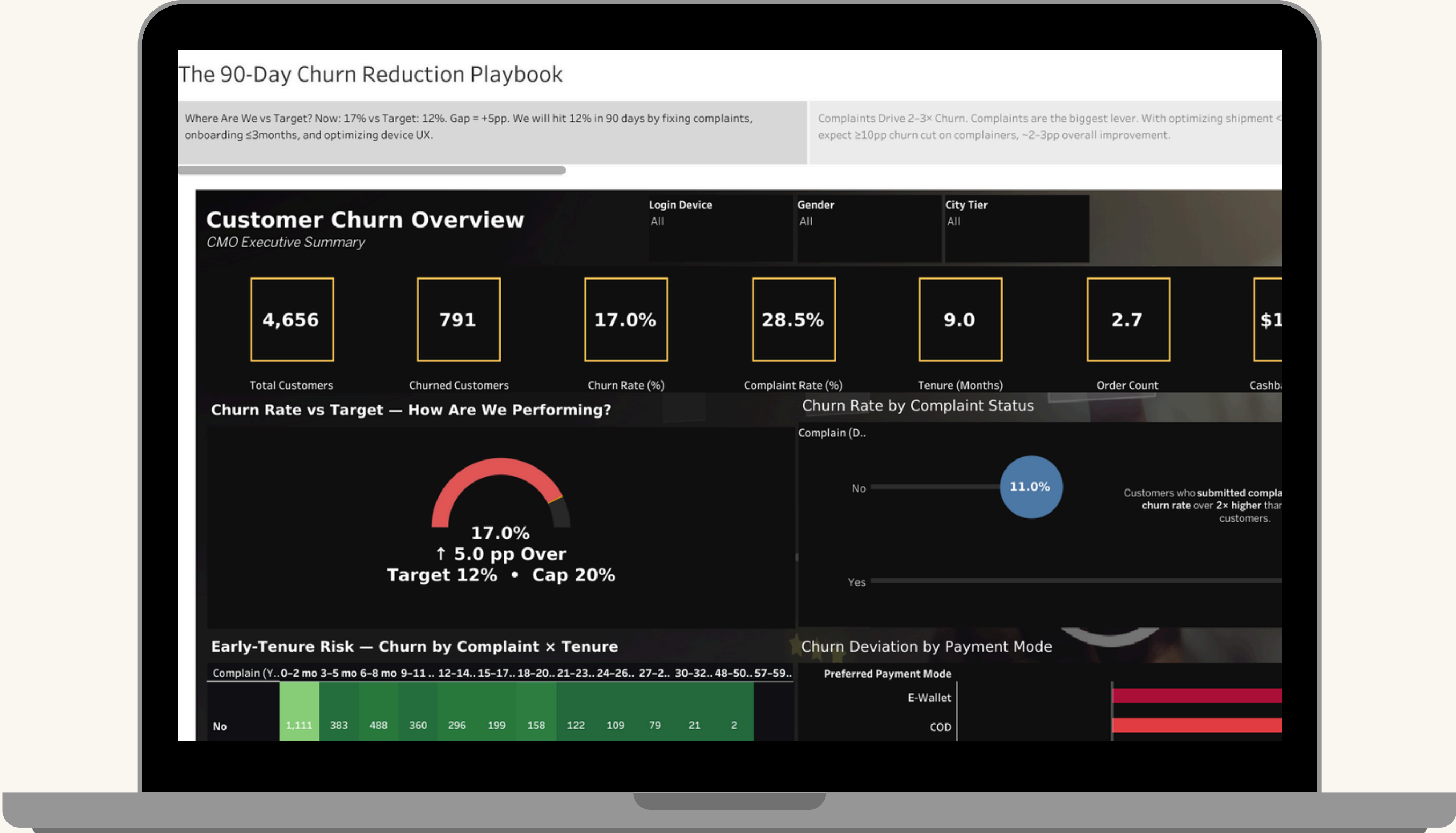
Recalibrate Threshold Periodically

Periodic recalibration ensures that the **trade-off between precision and recall** remains **aligned with current business goals**. Retrain the model with **newest data** to keep the model relevant to customer behavior.

More Feature Analysis

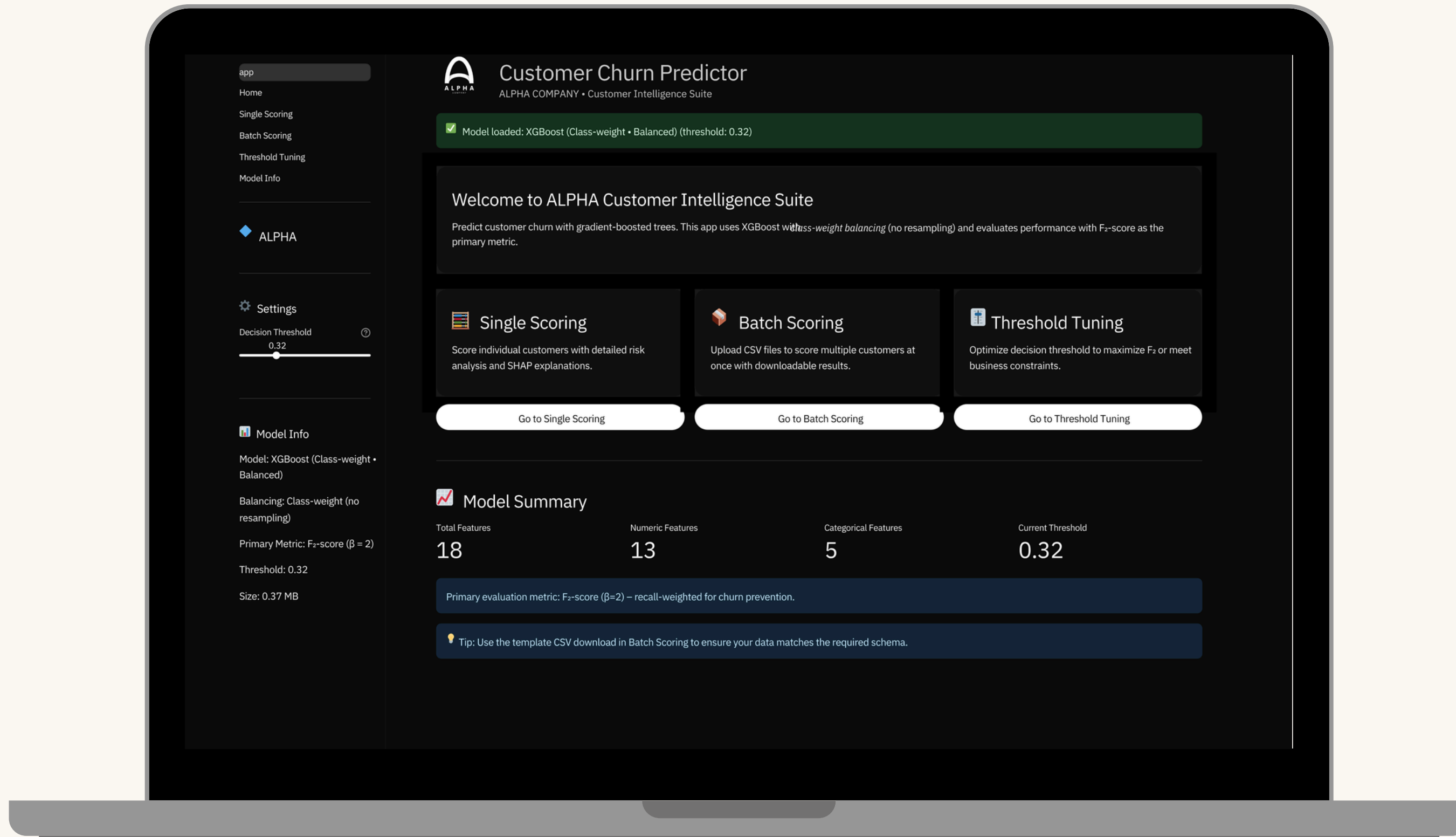
Use **advanced feature analysis** (e.g. Partial Dependence Plots (PDP) & Individual Conditional Expectation (ICE)) to show the average **effect of one or two features on churn probability & revealing non-linear or segment-specific behaviors**.

Tableau Dashboards



Interactive Dashboards: [Visit Alpha Churn Reduction Playbook](#)

Streamlit App Deployment



Live App: [Visit Alpha Churn Predictor](#)

THANK YOU!

Let's Stay in Touch:



Ardinata J.K Tambun

 [linkedin.com/in/ardinatatambun/](https://www.linkedin.com/in/ardinatatambun/)

 github.com/ardinatatambun/



Alfriando C Vean

 [linkedin.com/in/alfriandocvean/](https://www.linkedin.com/in/alfriandocvean/)

 github.com/alfcvean/



Bonifasius Sinurat

 [linkedin.com/in/bonifasiusx/](https://www.linkedin.com/in/bonifasiusx/)

 github.com/bonifasiusx/